

# IS THERE A PLATONIC TREE?

## CALIBRATING LATENT HYPERBOLICITY

### IN FOUNDATION MODELS

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#### ABSTRACT

Hyperbolic representation learning rests on a premise we call latent hyperbolicity: standard models are already tree-like because their class geometry scores a low Gromov  $\delta$ , a measure of tree-likeness that is zero for a tree. That low value is also what a random cloud of the same dimension and spread scores, so we build an instrument that reads every score as its excess over many such random clouds, and add a test for hierarchy whose ability to detect one is measured. We apply it to 12 vision backbones, 6 datasets and 15 text models. On image features, where the premise is read, the calibrated reading is indistinguishable from a random cloud in most model–dataset cells and small elsewhere. Class centroids do carry structure that a random cloud of the same shape does not, in 44 of 72 cells, but so does a star of clusters. In 4 of 12 ImageNet backbones the test finds structure, but it is the orientation of each cluster relative to the centers of the superclasses, not a hierarchy among those centers: randomizing the orientations removes it, whereas a hierarchy planted among the centers, synthetic or trained in, survives. In the nine backbones where such a planted hierarchy is detected, none is found in the real model; in the other three the test cannot tell. A backbone trained in hyperbolic space, MERU, shows the same structure as its Euclidean twin and stays so close to the origin that its space is nearly flat. Across models, the usual way of comparing trees makes the self-supervised models look like outliers, an artifact of how the trees are cut into groups; compared properly, the trees agree on which classes group together almost as well as two readings of the same model do, but not on the distances between them; and the self-supervised models organize classes by direction rather than by distance. Read correctly, foundation models organize classes into clusters, agree on how those clusters nest almost as much as noise allows but not on the distances between them, show no hierarchy among the superclasses where the test can see one, and their raw tree-likeness is not evidence for hyperbolic geometry.

## 1 INTRODUCTION

Semantic knowledge is hierarchical, and hyperbolic space is the natural geometric home for hierarchies (Nickel & Kiela, 2017; Sarkar, 2011; Ganea et al., 2018; Chami et al., 2019). A line of work therefore imposes it on representations at training time, from vision to vision–language and language (Khurlov et al., 2020; Ermolov et al., 2022; Atigh et al., 2022; Bdeir et al., 2024; Desai et al., 2023; Pal et al., 2025; Chen et al., 2022; He et al., 2025; Sinha et al., 2024). These methods cite a premise, which we call *latent hyperbolicity*: standard models already organize concepts in a tree-like metric structure. The evidence is a single number, the Gromov  $\delta$  of their features, which is zero for a metric tree and grows as a metric departs from one. Measured on CNN and ViT features (Khurlov et al., 2020; Bdeir et al., 2024), on token embeddings (Yang et al., 2025) and on word vectors (Tifrea et al., 2019), it comes out low. Those readings are taken at the sample level, on individual images or tokens rather than on classes, and they are never calibrated: a value is called low on its own, without asking what a structureless cloud of the same dimension and spectrum would score. Is the premise right, and what does it license? Figure 1 frames the question: an observer who sees only a shadow, the raw  $\delta$ , asks which of three worlds cast it, a structureless cloud, a star of clusters or a tree. All three read alike.

Why is the reading low? Gromov  $\delta$  assigns zero to trees (Gromov, 1987), but its reference level depends on dimension, spectrum and statistic, so a raw value cannot be called low on its own. High-dimensional clouds concentrate toward equidistance (Beyer et al., 1999; Aggarwal et al., 2001),

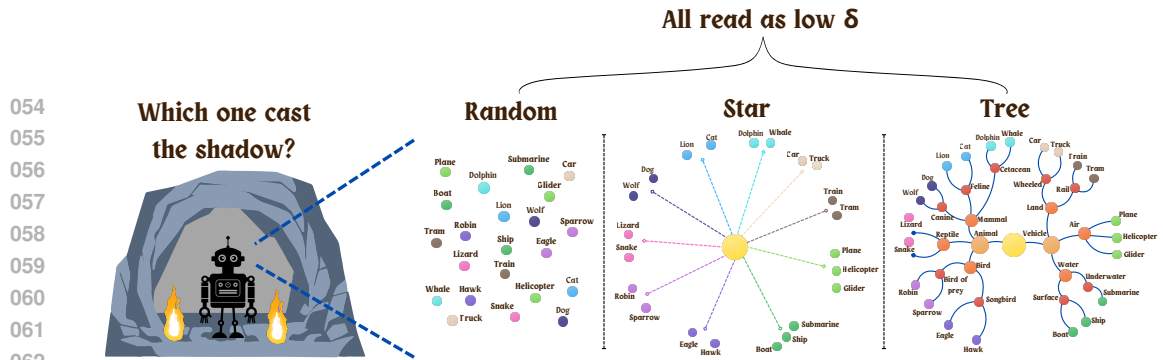


Figure 1: **Plato’s cave: a random cloud, a star and a tree cast the same shadow.** An observer who sees only shadows cannot tell which object cast them, and a raw  $\delta$  is such a shadow: points with no structure, clusters around a hub, and a nested hierarchy all read as low  $\delta$ . The instrument reads the objects instead of the shadow: no structure beyond the null in the cloud, clusters without depth in the star, clusters with depth in the tree.

which is a degenerate star tree: we call this the *dimension confound*, the geometric counterpart of the width confounder of Gröger et al. (2026). Anisotropic spectra lower the effective dimension and raise the reading: the *spectrum confound*. And the supremum over sampled quadruples grows with the budget and does not converge (Fournier et al., 2015): the *statistic confound*. The three are visible in real models: GPT-2 M sits at its matched null despite a low raw reading, and classic sentence embedders share their raw reading with vision-contrastive models while carrying none of their structure.

We build the instrument that reads  $\delta$  correctly. It calibrates the estimator on reference geometries, expresses every reading as an *excess* over a null cloud matched in dimension and covariance spectrum, ranks that excess against matched replicates, and adds a *depth test* whose false-alarm rate and power are measured on real clouds. With it we ask of geometry what convergence work asks of similarity (Huh et al., 2024; Gröger et al., 2026): what survives calibration, and is it shared? They calibrate similarity across models; we calibrate geometry within a model, and comparing trees across models needs a calibration of its own.

**The answer has three parts.** On image features, where the premise is read, the calibrated reading is indistinguishable from a random cloud of the same shape in most cells and small elsewhere; the four values reported by Khrulkov et al. (2020) are reproduced and, calibrated, two of them are indistinguishable from a random cloud. On class centroids there is structure that a random cloud does not have, in 44 of 72 cells and 30 of 36 on datasets with 47 classes or more, but a star of clusters already produces it. The depth test certifies in 4 of 12 ImageNet backbones that clusters are oriented toward the superclass centers, which we call hubs, not that the hubs form a hierarchy within the grouping of classes into superclasses, which we call the frame: randomizing the orientations removes every verdict, whereas an implanted hierarchy, synthetic or trained in, survives it; no hub hierarchy is found in the nine backbones where an implanted one is detected once cluster orientations are randomized, and the test is blind in the other three. A backbone trained in hyperbolic space shows the same structure as its Euclidean twin and lives in the near-flat regime. Across models, the usual way of comparing trees makes the self-supervised models look like outliers, an artifact of how the trees are cut into groups; compared properly, the trees agree on which classes group together almost as well as two readings of the same model do, but not on the distances between them; and the self-supervised models organize classes by direction rather than by distance.

#### Our contributions are:

1. **The instrument.** Estimator calibration, spectrum-matched nulls read as corrected ranks, and a depth test with measured false-alarm rate and power.
2. **The census.** Structure beyond a random cloud in most models, clusters oriented toward their hubs in a few, no hub hierarchy in the nine backbones where an implanted one is detected once cluster orientations are randomized, a blind test in the other three, and a nominally hyperbolic backbone as the control for imposing the geometry.
3. **The map of trees.** The island is an artifact of the cut; the trees agree on which classes group together almost as well as two readings of the same model do, but not on the distances between them, and the self-supervised tree is angular.
4. **Consequences for practice.** What to measure before imposing curvature, and which zero-cost readout collects what is there.

## 2 RELATED WORK

Reading  $\delta$  against a matched reference has precedent in network science, where the  $\delta$  of real graphs is compared with that of random graphs of comparable size (Narayan & Sanjeev, 2011; Kennedy et al., 2013; Adcock et al., 2013); we bring the idea to embedding clouds, where the reference must match dimension and spectrum. The curvature itself has been studied as a representation tradeoff (Sala et al., 2018) and as a mixed-curvature product to be learned (Gu et al., 2019), which presupposes a diagnostic of the kind we calibrate. The analyses of learned representations characterize Euclidean structure and do not test for tree-likeness: representation similarity, intrinsic dimension and neural collapse (Kornblith et al., 2019; Ansuini et al., 2019; Pope et al., 2021; Pappas et al., 2020). Closest to us is the work on representational convergence: the Platonic Representation Hypothesis (Huh et al., 2024) argues that models converge toward shared representations, and its cross-modal convergence has been stress-tested at scale (Koeperke et al., 2026); Gröger et al. (2026) show that permutation-calibrated similarity removes most of that convergence while local neighborhood structure survives, and Park et al. (2024) read concept hierarchy in language models from the orthogonality of concept directions. We are the geometric complement of this program: the same calibration applied within a model instead of across models, with a depth test measured on real clouds, a hyperbolic backbone as the control for imposing the geometry, and a calibration of the tree-to-tree comparison itself.

## 3 METHODOLOGY

### 3.1 BACKGROUND: GROMOV $\delta$

Tree-likeness is measured on quadruples. Take any four points  $w, x, y, z$  of a metric space with distance  $d$ . They can be paired in three ways, and each pairing has a sum of two distances:

$$d(w, x) + d(y, z), \quad d(w, y) + d(x, z), \quad d(w, z) + d(x, y). \quad (1)$$

Sort the three sums as  $S_{\max} \geq S_{\text{mid}} \geq S_{\min}$ . In a tree the two largest sums are always equal, because the two pairings that cross the trunk traverse the same path, so the four-point defect  $(S_{\max} - S_{\text{mid}})/2$  is zero; the further a metric is from a tree, the larger the defect can become. A metric tree has  $\delta = 0$ . Euclidean space has no bound at all: enlarging a square enlarges its defect in proportion. Hyperbolic space sits in between, with a bounded defect whose value for the hyperbolic plane is known,  $\ln 2$  under this definition (Nica & Špakula, 2016).

**Definition 1** (four-point defect). For a quadruple  $(w, x, y, z)$  with the pairing sums of Eq. 1 sorted as  $S_{\max} \geq S_{\text{mid}} \geq S_{\min}$ , the four-point defect is  $\delta(w, x, y, z) = \frac{1}{2}(S_{\max} - S_{\text{mid}})$ , and the Gromov  $\delta$  of a metric space is the supremum of the defect over its quadruples (Gromov, 1987).

### 3.2 ESTIMATION AND NORMALIZATION

Three steps turn  $\delta$  into a number that can be compared across models. First, a thousand centroids have far more quadruples than can be enumerated, so the defect is computed on half a million random quadruples. Second, the largest defect in that sample is a poor estimate of the supremum: it belongs to a single extreme quadruple and keeps growing as more quadruples are drawn (Fournier et al., 2015). We therefore read the 99.9th percentile of the sampled defects,  $\hat{\delta}_{99.9}$ , the value that 99.9 per cent of them fall below: it keeps the tail of the defects, as the supremum does, but it is set by hundreds of quadruples rather than by one. It also does not move with the sample size: across quadruple budgets the ImageNet reading drifts by less than half its own spread (Table 3). Third, the defect grows with the size of the point set, so we divide by the diameter, the largest pairwise distance:

$$\delta_{\text{norm}} = \frac{\hat{\delta}_{99.9}}{\text{diam}}. \quad (2)$$

On this scale a tree reads zero, and a hyperbolic region of radius four or more reads below a sphere (Table 4).

**Definition 2** (reading of record). The reading of record of a cloud is  $\delta_{\text{norm}}$  of Eq. 2, with  $\hat{\delta}_{99.9}$  computed over half a million sampled quadruples per seed and averaged over ten seeds.

**Proposition 1** (range bound and dimension confound). Let  $d_{\min}$  and  $d_{\max}$  be the smallest and the largest pairwise distance of a finite metric space. (a) For any quadruple the four-point defect is at most  $d_{\max} - d_{\min}$ , hence  $\delta_{\text{norm}} \leq 1 - d_{\min}/d_{\max}$ . (b) For an iid Gaussian cloud of  $n$  points in dimension  $d$ , with  $n$  fixed,  $d_{\min}/d_{\max} \rightarrow 1$  as  $d$  grows (Beyer et al., 1999; Aggarwal et al., 2001), so  $\delta_{\text{norm}} \rightarrow 0$  without any structure.



Figure 2: **The raw reading cannot be read on its own.** (a) Raw reading and matched null per ImageNet backbone, ordered by dimension; the gap is the excess. (b) Same dataset, same reading, opposite verdict: ViT-B on CIFAR-100 images and DINO-B on CIFAR-100 centroids.

**A raw value cannot be called low on its own.** The reference level of the reading depends on dimension, spectrum and statistic. High dimension lowers it (the dimension confound), as Proposition 1 states for a structureless cloud. Anisotropy raises it through the effective dimension (the spectrum confound), and a sampled supremum grows with the budget (the statistic confound); the calibration on reference geometries confirms this (Table 4).

### 3.3 SPECTRUM-MATCHED NULL, EXCESS AND GENUINENESS

**Definition 3** (Haar null). Let  $X = U\Sigma V^\top$  be the singular value decomposition of the centered cloud. A null replicate is  $X' = Q\Sigma V^\top$  with  $Q$  the orthogonal factor of a column-centered Gaussian matrix, orthogonal to the all-ones vector and Haar-distributed on the centered subspace, so  $X'$  has exactly the centered singular values of  $X$  and structureless coefficients.

**The null is a cloud with the real shape and no structure.** Gaussian coefficients reproduce the spectrum only in expectation and inflate the excess of low-rank clouds. The centered Haar construction reproduces the centered spectrum exactly, so it is the record. The uncentered and the Gaussian readings are reported beside it (Table 5).

**Definition 4** (excess). With  $K$  null replicates  $X'_1, \dots, X'_K$  of Definition 3, the excess of a cell is

$$e(X) = \delta_{\text{norm}}(X) - \frac{1}{K} \sum_{k=1}^K \delta_{\text{norm}}(X'_k). \quad (3)$$

**Size and evidence.** The excess is a size; the evidence is the rank of the reading among its replicates and the left-tail  $p$  that the rank gives,

$$r = \#\{k : \delta_{\text{norm}}(X'_k) > \delta_{\text{norm}}(X)\}, \quad p = \frac{1 + K - r}{K + 1}, \quad K = 200. \quad (4)$$

**Definition 5** (genuine). Order the  $m$  left-tail  $p$ -values of a census as  $p_{(1)} \leq \dots \leq p_{(m)}$ . A model-dataset cell is genuine when its  $p$  is at or below the Benjamini-Hochberg threshold at level five per cent,

$$p \leq p_{(i^*)}, \quad i^* = \max\{i : p_{(i)} \leq 0.05 i/m\}, \quad (5)$$

and the word is used in this sense only.

**What a genuine excess means.** A negative genuine excess certifies structure beyond the second moments; clustering is its most plausible reading, supported by the superclass recovery of Section 5.4. Two cells on the same dataset with the same reading can receive opposite verdicts (Figure 2).

### 3.4 DEPTH TEST

**Definition 6** (hub null and excess B). Given a frame that assigns each centroid  $x_i$  to a cluster  $k(i)$  with hub  $h_{k(i)}$ , the cluster mean, write  $x_i = h_{k(i)} + o_i$ . The hub null keeps every offset  $o_i$  and replaces the hubs by a Haar resample with their spectrum,  $x'_i = h'_{k(i)} + o_i$ , and the excess of Eq. 3 against this null,  $e_B$ , measures structure above the clusters of the frame only.

**Definition 7** (matched star). The matched star of a cloud keeps the real clusters, each as a Haar sample with the cluster's own covariance, and places their hubs as Gaussian draws at the real hub radius: the clustering of the real cloud with no hierarchy above it. Its spectrum-matched-hub variant draws the hubs as a Haar resample of the real hubs.

**Definition 8** (decoupling control). The decoupling control keeps the hubs of the frame and rotates each cluster’s offsets by an independent Haar rotation. A verdict that survives the decoupling is carried by the arrangement of the hubs; one that does not is carried by the orientation of the clusters relative to their hubs, which we call alignment.

**The depth test compares the real cloud with its matched star.** With  $e_B$  read on the real cloud and on its matched star, the depth statistic and its standardized form are

$$\text{depth} = e_B(X) - e_B(X_{\text{star}}), \quad z = \text{depth} / \text{s.d.}, \quad (6)$$

The spread is taken over star seeds and null replicates, and a cloud is certified against its matched star when  $z \leq -2$ . The power of the test is the fraction of implanted hierarchies it detects and its false-alarm rate the fraction of flat controls that fire; both are measured on real clouds in Section 5.3.

### 3.5 COMPARING TREES ACROSS MODELS

**Trees are compared under a configuration chosen before the comparison.** Two hierarchies are compared by the adjusted Rand index between their cut partitions, by the correlation of their cophenetic distances, and by their agreement on which pair of a random class triplet merges first. The configuration of metric and linkage is selected per dataset by a criterion fixed without reference to the agreements: the highest cophenetic fidelity among configurations whose reference cuts are non-degenerate for every model. Local agreement is read with the permutation calibration of Gröger et al. (2026) on mutual-kNN agreement and centroid CKA.

### 3.6 PROJECTION AND TASKS

**Three readouts are compared on the intact representation.** A parameter-free map centers the features, scales the 95th-percentile norm to radius  $1/\sqrt{2}$  and preserves directions. Euclidean, cosine and Poincaré readouts are then scored by nearest-centroid classification and 5-way 5-shot episodes on every cell of the task grid.

## 4 EXPERIMENTAL SETUP

**The census has twelve backbones, six class sets and fifteen text models.** The vision backbones are four supervised ViTs trained on ImageNet-21k, DINO-B, four DINOv2 sizes, CLIP-B, CLIP-L and SigLIP-B. The class sets are ImageNet and CIFAR-100, which carry a real hierarchy, CIFAR-10 and DTD, and the flat FMNIST and MNIST. The text models read the thousand ImageNet class names: four GPT-2 sizes, three Pythia sizes, OLMo-1B and seven sentence embedders (Table 6).

**Four controls test the alternatives.** Two self-supervised ResNets read the vision census. DeiT-B and the augreg ViT-B, trained on ImageNet-1k leaf labels, test whether a hierarchical label set explains the depth. MERU and its Euclidean CLIP twin test whether training in hyperbolic space creates depth. The census ViT-B fine-tuned with and without a hierarchical objective is a trained control.

**The premise is read on images and hierarchy on centroids.** The sample-level reading uses about a thousand stratified training images per cell on CIFAR-100 and DTD. Class centroids average a hundred cached training images per class on ImageNet and the full cached split elsewhere. Causal language models give the last-token state of “a photo of a {class}”, extracted one prompt at a time because batching alters GPT-2’s hidden states, and embedders use their recommended pooling (Table 7).

**Calibration budget and frames.** Each cell is read against two hundred Haar replicates, with half a million sampled quadruples per seed and ten seeds. With ten classes there are only 210 quadruples, so  $\hat{\delta}_{99.9}$  coincides with the supremum there. Benjamini–Hochberg runs over the vision census and, separately, over the text census. The depth test uses the WordNet cut into thirty superclasses on ImageNet, the twenty coarse labels on CIFAR-100, a balanced frame as a control, and ten star seeds.

## 5 RESULTS

### 5.1 THE PREMISE AT THE SAMPLE LEVEL

**Raw readings are not evidence, ours or published, and calibrated ones are weak and model-dependent.** On per-image features, where the premise is read, the calibrated reading is not genuine in most cells. Where it is genuine, the excess removes at most two fifths of the null reading. Table 8 gives the reading per cell.

**A published reading is reproduced and calibrated.** Their estimator on our extraction reproduces the raw  $\delta_{\text{rel}}$  that Khrulkov et al. (2020) report for ResNet-34 on four datasets within 0.03. Calibrated on the same clouds, the excess is negative on every dataset, and only CIFAR-100 and MiniImageNet fall below the null in every batch at the 0.05 level. Table 1 gives the four rows.

| dataset      | their $\delta_{rel}$ | ours, their estimator | excess  | $r/200$ , largest $p$ |
|--------------|----------------------|-----------------------|---------|-----------------------|
| CIFAR-10     | 0.26                 | 0.275                 | -0.0037 | 173/200, 0.333        |
| CIFAR-100    | 0.25                 | 0.261                 | -0.0087 | 198/200, 0.030        |
| CUB-200      | 0.25                 | 0.270                 | -0.0043 | 177/200, 0.249        |
| MiniImageNet | 0.21                 | 0.203                 | -0.0234 | 200/200, 0.005        |

Table 1: **A published reading reproduced and calibrated.** ResNet-34 setting of Khurlov et al. (2020): their relative hyperbolicity, ours with their estimator, the excess, and the rank of 200 with the largest left-tail  $p$ .

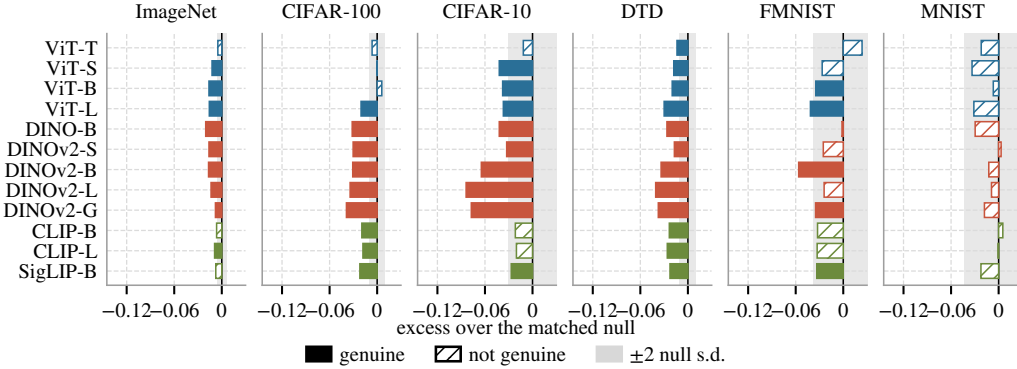


Figure 3: **Most cells show structure beyond the second moments.** Excess of the reading of record over its matched null per dataset and backbone, filled when genuine; band: two null standard deviations.

## 5.2 STRUCTURE BEYOND THE SECOND MOMENTS AT THE CLASS LEVEL

**Most cells show structure beyond the second moments.** It is genuine in 44 of 72 cells and 30 of 36 on the three datasets with 47 classes or more. On ImageNet and CIFAR-100 the count is 18 of 24. Figure 3 and Table 2 give the census.

**The count survives resampling.** Folding image resampling and the estimator’s seeds into the null spread, or repeating the census on resampled centroids, keeps most of the count. Table 3 gives the bootstrap.

**No family owns the structure.** The smallest backbone is genuine on one dataset only, and the flat datasets have smaller and less consistent excesses; FMNIST is genuine in 5 of 12 backbones. A class-count control shows that the excess shrinks with the number of classes, so only the verdict carries across datasets. Tables 5 and 3 give the details.

**A star of clusters already passes the census.** A star of clusters has structure beyond second moments. A synthetic star with no hierarchy scores an excess as deep as any real cell, so a negative excess says structure beyond the second moments, not depth. Table 4 gives the star’s excess.

**Neural collapse is the flat limit, not what the census sees.** Neural collapse predicts that supervised class means converge to a simplex frame (Papayan et al., 2020), a star with no hierarchy. The census does not certify that frame: class means cluster around superclass hubs, which the superclass recovery of Section 5.4 shows and the depth test reads as hub alignment.

## 5.3 DEPTH ABOVE THE LABELLED CLUSTERS

**The depth test certifies that clusters are oriented toward their hubs in four backbones.** It certifies 4 of 12 backbones, ViT-S, ViT-B, ViT-L and DINOv2-L, under both matched stars. The noise level of a cloud is its within-cluster spread relative to the distance between its hubs. Figure 4a shows the  $z$  per backbone and Table 9 gives both stars.

**What it certifies is alignment.** Under the decoupling control none of the four fires, and ViT-L reads -1.61, within the -1.61 to -1.76 of the flat control: without its orientations the real cloud reads like a flat one. The alignment is relational, needing both the real hubs and the orientation of each cluster relative to them: randomizing either removes it, 0 of 60 with random hubs and 0 of 4 with random orientations. The alignment survives removing the radial component of every offset in all four certified backbones,  $z$  from -2.26 to -3.99 under both stars, which excludes the spread of feature norms as its source. Under full L2 normalization, which also moves the hubs, it survives in ViT-B and ViT-L only. The implanted principal-axis alignment reproduces the verdict in

| model    | family | $\hat{\delta}_{99.9}$ IN | excess IN      | $r/200$ ( $p$ ) | exc. C100      | exc. C10       | $\rho_{WN}$ |
|----------|--------|--------------------------|----------------|-----------------|----------------|----------------|-------------|
| ViT-T    | sup.   | 0.088                    | $-0.005^\circ$ | 169 (.159)      | $-0.006^\circ$ | $-0.012^\circ$ | +0.52       |
| ViT-S    | sup.   | 0.063                    | $-0.012$       | 200 (.005)      | $-0.000^\circ$ | $-0.042$       | +0.50       |
| ViT-B    | sup.   | 0.054                    | $-0.016$       | 200 (.005)      | $+0.006^\circ$ | $-0.038$       | +0.49       |
| ViT-L    | sup.   | 0.055                    | $-0.016$       | 200 (.005)      | $-0.020$       | $-0.037$       | +0.53       |
| DINO-B   | SSL    | 0.055                    | $-0.020$       | 200 (.005)      | $-0.031$       | $-0.042$       | +0.36       |
| DINOv2-S | SSL    | 0.053                    | $-0.016$       | 200 (.005)      | $-0.030$       | $-0.033$       | +0.22       |
| DINOv2-B | SSL    | 0.034                    | $-0.017$       | 200 (.005)      | $-0.031$       | $-0.065$       | +0.18       |
| DINOv2-L | SSL    | 0.030                    | $-0.014$       | 200 (.005)      | $-0.034$       | $-0.084$       | +0.18       |
| DINOv2-G | SSL    | 0.033                    | $-0.008$       | 200 (.005)      | $-0.039$       | $-0.077$       | +0.20       |
| CLIP-B   | contr. | 0.084                    | $-0.007^\circ$ | 172 (.144)      | $-0.019$       | $-0.022^\circ$ | +0.57       |
| CLIP-L   | contr. | 0.076                    | $-0.009$       | 197 (.020)      | $-0.018$       | $-0.020^\circ$ | +0.59       |
| SigLIP-B | contr. | 0.075                    | $-0.008^\circ$ | 189 (.060)      | $-0.022$       | $-0.027$       | +0.58       |

Table 2: **Structure beyond the second moments is the rule at the class level.** Per backbone: raw ImageNet reading  $\hat{\delta}_{99.9}$ , excess over the Haar null mean, rank  $r/200$  with left-tail  $p$ , genuine = BH  $p \leq 0.05$  over 72 cells ( $^\circ$ : not genuine; bold: above the null), the CIFAR-100 and CIFAR-10 excess, and  $\rho_{WN}$ , the Spearman correlation with WordNet distances.

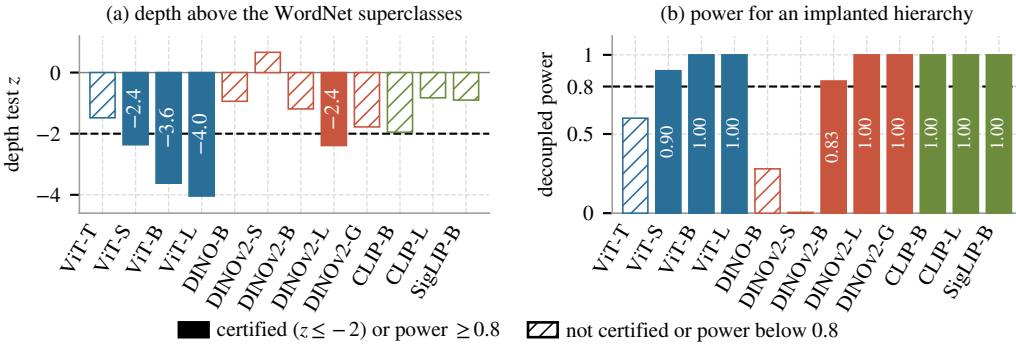


Figure 4: **The depth test certifies in 4 of 12 ImageNet backbones that clusters are oriented toward their hubs, not that the hubs of the frame form a hierarchy; no hub hierarchy is found in the nine backbones where the decoupled control detects an implanted one, and the test is blind in the other three.** (a) Depth  $z$ , filled when certified; line at  $z = -2$ . (b) Decoupled power for an implanted three-level hierarchy, filled at 0.8 or more; line at 0.8.

the supervised ViTs and CLIP-B but not in DINOv2-L, and its exact geometric form is not resolved here.

**What the test can see.** The test misses an implanted two-level tree at this noise level but detects a three-level hierarchy with ViT-L’s spectrum and noise in 4 of 5 seeds, and that detection survives orientation randomization, whereas none of the four real verdicts does. The decoupling control fires on every deep seed and reads deeper than the intact cloud. The deeper reading comes from both sides: decoupling shrinks the star spread and deepens the excess below the star. The decoupled control detects the same hierarchy, built at each backbone’s own spectrum and ratio, in nine backbones (decoupled power 0.83–1.00) and not in the other three (power 0.01–0.60). The three blind backbones sit at ratios 1.5 to 3.0, inside the 1.3 to 3.9 of the nine covered, so the power follows the backbone, not its ratio or family. ViT-B fine-tuned with a hierarchical cross-entropy at three WordNet cuts keeps firing once decoupled, 10 of 10, whereas the leaf-CE and frozen models do not, 0 of 10, on the frame of record. The pre-set criterion asked the leaf-CE and frozen models not to be certified intact, which they are, by alignment, so the discriminating comparison is the decoupled one. A flat control with the same spectrum and ratio as the deep one fires in 0 of 5 intact and in 8 of 50 decoupled runs. The decoupled test is therefore biased toward firing, so the four real backbones not firing once decoupled is conservative evidence, and the synthetic hierarchy at  $z = -4.19$  to  $-4.82$  stands well clear of that bias. On the balanced frame the matched flat control fires in 0 of 50 decoupled runs. The WordNet Poincaré embeddings (Nickel & Kiela, 2017), read with Euclidean distances, do not fire. Tables 10 and 9 give the runs.

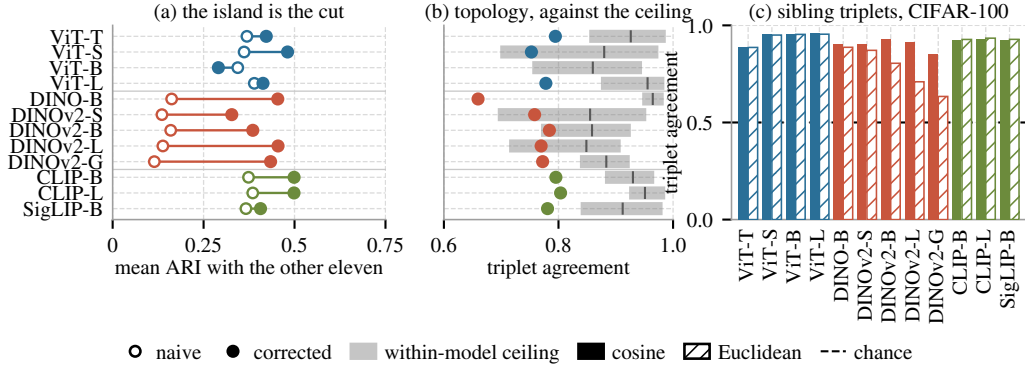


Figure 5: **The island is the cut; tree topology is shared almost up to noise; the self-supervised structure is angular.** (a) Mean agreement of each model’s tree with the other eleven, naive (hollow) and corrected (filled). (b) Triplet agreement under the corrected comparison against the within-model ceiling (band). (c) Sibling-triplet agreement under cosine and Euclidean distance.

**No hub hierarchy is found where the test has power.** The power is the decoupled control’s: it detects an implanted hierarchy in ViT-S/B/L, DINOv2-B/L/G, CLIP-B/L and SigLIP-B, and the test is blind in ViT-T, DINO-B and DINOv2-S. The decoupling separates structure among the hubs of the frame from structure carried by cluster orientations. A hierarchy below the frame expressed in how clusters open would be removed with the orientations and counted as alignment, so no hub hierarchy means no hierarchy among the frame’s hubs. Figure 6 gives the two-level detection rates and Table 10 the power per backbone.

**The certified set depends on the frame.** Under a balanced frame ViT-T joins the certified set and ViT-S and DINOv2-L leave it, and other WordNet cuts change the set again. Only ViT-B and ViT-L are certified under every frame. A flat cloud clustered under the frame of record fires under the balanced frame in 38 of 50 decoupled runs, so a real cloud read under a frame that is not its own is expected to fire. The frozen ViT-B firing in 7 of 10 there is consistent with that mismatch and is not evidence of hub structure.

**Leaf labels can produce the alignment but do not guarantee it.** The augreg ViT-B trained on ImageNet-1k leaf labels is certified and aligned with WordNet like the ImageNet-21k ViTs, whereas DeiT-B, trained on the same labels with another recipe, is neither. A hierarchical label set therefore does not explain the certified ViTs’ depth. Table 9 gives the two models.

**Imposing the geometry does not create hub structure.** MERU embeds images on the Lorentz hyperboloid with an entailment objective, and its Euclidean twin shares backbone, data and recipe (Desai et al., 2023). MERU shows the same clustering, no such structure either, and embeddings that never leave the near-flat regime. Table 9 gives the readings and the radii.

#### 5.4 WHOSE TREE

**The naive map manufactures an island.** Figure 5 compares the trees of the twelve models: (a) as usually compared, (b) under the criterion of Section 3.5 against the within-model ceiling, and (c) by whether classes are grouped by direction or by distance. Under Euclidean distances with average linkage the DINOv2 family agrees with the supervised and contrastive block far less than the block agrees with itself. The cause is chaining: DINOv2’s Euclidean dendrograms put almost every class in one cluster, forcing the index to zero.

**Controlling the cut leaves a moderate gap.** Under the selected configuration, cosine distances with average linkage on ImageNet, the island shrinks to a gap beyond resampling noise. Two resamples of the same model agree on 0.85 to 0.96 of the triplets across backbones, mean 0.90, against 0.77 for DINOv2 against the block and 0.76 within it. The cophenetic correlation reaches 0.91 to 0.97 within a model, mean 0.94, against 0.48 and 0.78 across models. The ARI at the cut reaches 0.64 to 0.94, mean 0.79, against 0.39 and 0.48. The tree topology is therefore shared to within most of the ceiling and the metric agrees at about half of it. About a third of the within-block agreement is missing under the admissible configurations as a whole. Figure 5b and Table 11 give the map, the configurations and the ceiling.

**The self-supervised tree is angular.** DINOv2’s semantics concentrate in the angular component: sibling triplets are resolved far better under cosine than under Euclidean distance, so the Euclidean dendrograms chain. The cosine census agrees with the Euclidean one in most cells, and DINOv2 on ImageNet is genuine under both. Figure 5c shows the triplets and Table 5 gives the cosine census.

**Alignment with WordNet follows supervision and recipe.** The correlation between inter-centroid and WordNet distances is highest for the contrastive VLMs, then the supervised ViTs, and lowest for DINOv2. Supervision deepens the resemblance on average, and DeiT-B shows that the effect is recipe-dependent. Table 12 gives the correlations.

**The recovery is not WordNet circularity.** On DBpedia Classes, an ontology independent of WordNet, the three text embedders read a genuine excess and recover the true partition under every configuration. On HierarCaps, a four-level caption hierarchy with no class set (Alper & Averbuch-Elor, 2024), the encoders resolve sibling triplets far above chance. Table 12 gives both.

**Text depends on recipe and scale.** On the ImageNet class names, 7 of 15 text models are genuine under the record. GPT-2 S is genuine at the margin, GPT-2 M is not, and GPT-2 L and XL are genuine under every reading, as are Pythia at every scale and OLMo-1B. Table 7 gives the census.

## 5.5 WHAT IS SHARED IS LOCAL

**Models share neighborhoods, not metrics.** Permutation-calibrated mutual-kNN agreement and centroid CKA survive across all sixty-six model pairs. The Poincaré readout improves neither: models share who is near whom and differ in the metric (Gröger et al., 2026). Table 13 gives the values.

## 6 IMPLICATIONS FOR HYPERBOLIC REPRESENTATION LEARNING

**The raw reading cannot select a curvature.** Khrulkov et al. (2020) set the curvature of the Poincaré ball from the relative hyperbolicity,  $c = (0.144/\delta_{\text{rel}})^2$  with  $\delta_{\text{rel}} = 2\delta/\text{diam}$ . The rule, calibrated with the supremum, assigns structureless clouds on the Gaussian band, itself the sampled supremum rather than the census percentile, a curvature from 0.48 to 2.5, tracking the width of the backbone. Table 4 gives the band.

**Both readings predict the gain, and the depth verdict predicts none.** Within each hierarchical dataset the raw reading predicts the Poincaré readout’s gain as well as the calibrated excess does. The depth verdict predicts no hyperbolic gain. Calibration buys interpretation, a low  $\delta$  is not hierarchy, and not prediction. Table 14 gives both.

**The calibration certifies structure and does not choose the readout.** Before imposing curvature, certify that there is structure to justify it, then take the cheapest readout that collects it. Cosine collects the self-supervised structure, and the Poincaré readout adds +0.9 to +1.3 pp over cosine for the contrastive VLMs on prototype tasks and nothing consistent elsewhere. In few-shot learning a fixed-radius Euclidean encoder does at least as well as hyperbolic prototypes (Moreira et al., 2024).

## 7 CONCLUSION AND LIMITATIONS

**Conclusion.** Read correctly, foundation models organize classes into clusters, agree on how those clusters nest almost as much as noise allows but not on the distances between them, show no hierarchy among the superclasses where the test can see one, and their raw tree-likeness is not evidence for hyperbolic geometry. Curvature should be measured rather than assumed. A change of readout collects what is there.

**Limitations of the instrument.** (i) The excess certifies structure beyond second moments, star-like clustering included, so depth needs the separate test. (ii) The spectrum null conditions on the second moments, to which a hierarchy contributes, so a low excess does not exclude hierarchy carried by the spectrum. (iii) The depth test’s power for a three-level hierarchy follows the backbone rather than its within/between ratio or family. Under the decoupled control it is 0.83 or more in nine backbones and 0.60 or less in the other three. The verdict of no hub hierarchy therefore holds in the nine only, at 8 of 50 false alarms on flat clouds. It misses a two-level implant at this noise level, and the decoupled test is biased toward firing, which makes the real verdicts under decoupling conservative. It certifies alignment above its frame only: a hierarchy below the frame expressed in how clusters open would be removed with the orientations and counted as alignment, so no hub hierarchy means none among the frame’s hubs. Its trained positive control, one seed, discriminates the objectives only once decoupled. The alignment is not the radial spread of feature norms, which a control removes without changing the verdicts; full L2 normalization, a stronger transformation, keeps it in two of the four backbones. Implanted alignment is detected in 60% of runs at full strength, in every seed for the supervised ViTs and CLIP-B and in none for the DINOv2 family, so the power of the test for alignment is backbone-dependent and the DINOv2-L certification rests on a structure the implant does not reproduce. The alignment is thus relational, not radial, and reproduced by a principal-axis implant in the supervised ViTs and CLIP-B but not in DINOv2-L; its exact geometric form is not resolved here. (iv) The depth test is Euclidean, and for the angular DINOv2 tree it may be conservative. (v) A causal LM’s class representation depends on the extraction protocol as much

as on the model. (vi) The analysis rests on many choices, frames, stars and null variants among them, and a single pre-specified analysis is future work.

**Limitations of scope.** (vii) The census is class-centroid geometry on six datasets, and the sample-level reading covers two datasets, one image budget and one published setting. (viii) The objective-geometry link is correlational, resting chiefly on DINOv2’s scale range and on two leaf-label ViTs. (ix) Downstream scope is prototype-based metric tasks with modest gains.

#### REPRODUCIBILITY STATEMENT

Every number in the paper is annotated in the source with the result file that produces it, and all tables are generated by script from those files (Table 15). The full pipeline with fixed seeds will be released in the anonymized repository (`TODO(author)`). Its `tool/` directory ships the instrument as one script that reproduces any cell of Table 2 from its centroid matrix. Estimator settings are stated in Section 4 and the model panel in Table 6.

#### ETHICS STATEMENT

This work analyzes publicly released models and standard public datasets, involves no human subjects and no personal data, and proposes an analysis methodology and a metric-selection heuristic. We do not foresee ethical risks beyond those already associated with the underlying models and datasets.

#### AI USE STATEMENT

In this work, we used generative AI tools for writing assistance; for the retrieval of references; and for research ideation or execution, namely the implementation and execution of experiments under the authors’ direction and LLM-simulated reviews used as methodological feedback. We have not used generative AI tools for other tasks with required disclosure. We have reviewed all AI-assisted work, and we take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

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## A PROOFS

*Proof of Proposition 1.* (a) Each pairing sum in Eq. 1 adds two distances of the space, each of which lies in  $[d_{\min}, d_{\max}]$ . Hence  $S_{\max} \leq 2d_{\max}$  and  $S_{\min} \geq 2d_{\min}$ , so the defect  $\frac{1}{2}(S_{\max} - S_{\min})$  is at most  $d_{\max} - d_{\min}$  for every quadruple, and so is its supremum  $\delta$ . Dividing by the diameter  $d_{\max}$  gives  $\delta_{\text{norm}} \leq 1 - d_{\min}/d_{\max}$ ; the same bound holds for the 99.9th percentile of the sampled defects, which never exceeds their supremum. (b) Let  $x_1, \dots, x_n$  be iid standard Gaussian in  $\mathbb{R}^d$  with  $n$  fixed. For any pair  $i \neq j$  the coordinates of  $x_i - x_j$  are iid with variance 2, so  $\|x_i - x_j\|^2/(2d) \rightarrow 1$  almost surely as  $d \rightarrow \infty$  by the law of large numbers, and the convergence holds jointly over the finitely many pairs. Therefore  $d_{\min}/d_{\max} \rightarrow 1$ , and by part (a)  $\delta_{\text{norm}} \rightarrow 0$ . The rate is the distance-concentration rate of Beyer et al. (1999) and Aggarwal et al. (2001): the relative contrast  $(d_{\max} - d_{\min})/d_{\min}$  vanishes as  $d^{-1/2}$ . Part (b) is the dimension confound alone. The statistic confound is separate and elementary: a maximum over a growing sample of quadruples is non-decreasing in the sample size, so the sampled supremum can only rise with the budget.  $\square$

## B ONE TABLE PER QUESTION

Every table is generated by script from the released per-cell result files, and no value is hand-transcribed; the tables are numbered in the order the main text first cites them, one per question, with their result files indexed in Table 15.

### B.1 IS THE READING ROBUST? QUADRUPLE BUDGET, NULL CONSTRUCTION, RESAMPLING AND CLASS COUNT

(a) Quadruple budget under the record (Haar null, p99.9, 200 replicates): the excess of nine cells at  $10^4$  to  $2 \times 10^6$  sampled quadruples per seed; drift = range of the excess over budgets  $\geq 10^5$ , also divided by the excess's null spread.

| model    | data | $10^4$  | $5 \times 10^4$ | $10^5$  | $5 \times 10^5$ | $10^6$  | $2 \times 10^6$ | drift ( $\geq 10^5$ ) | drift/s.d. |
|----------|------|---------|-----------------|---------|-----------------|---------|-----------------|-----------------------|------------|
| ViT-L    | IN   | -0.0161 | -0.0161         | -0.0160 | -0.0157         | -0.0159 | -0.0159         | 0.0002                | 0.07       |
| DINOv2-L | IN   | -0.0131 | -0.0136         | -0.0137 | -0.0135         | -0.0136 | -0.0136         | 0.0002                | 0.38       |
| CLIP-B   | IN   | -0.0063 | -0.0060         | -0.0059 | -0.0065         | -0.0064 | -0.0064         | 0.0007                | 0.13       |
| ViT-L    | C100 | -0.0208 | -0.0208         | -0.0206 | -0.0205         | -0.0205 | -0.0205         | 0.0002                | 0.03       |
| DINOv2-L | C100 | -0.0348 | -0.0348         | -0.0348 | -0.0348         | -0.0347 | -0.0347         | 0.0001                | 0.07       |
| CLIP-B   | C100 | -0.0186 | -0.0195         | -0.0188 | -0.0195         | -0.0194 | -0.0194         | 0.0006                | 0.10       |
| ViT-L    | DTD  | -0.0305 | -0.0308         | -0.0312 | -0.0311         | -0.0313 | -0.0313         | 0.0002                | 0.05       |
| DINOv2-L | DTD  | -0.0418 | -0.0419         | -0.0421 | -0.0420         | -0.0422 | -0.0420         | 0.0002                | 0.06       |
| CLIP-B   | DTD  | -0.0243 | -0.0235         | -0.0242 | -0.0240         | -0.0243 | -0.0242         | 0.0003                | 0.05       |

Table 3: **The excess is budget-stable, the genuine count survives resampling and estimator noise, the excess shrinks with the class count, and training moves it.** (a) Excess against the quadruple budget for nine cells; drift = range of the record excess over budgets  $\geq 10^5$ , divided by the excess's spread in the last column; under the record every sign holds at every budget. (b) Bootstrap of the excess over resampled images per class on ImageNet, CIFAR-100 and DTD: the bootstrap s.d. is at most 0.0026 and every cell keeps the sign of its record excess in at least 0.67 of its resamples.

(b) Centroid bootstrap under the record (Haar null, p99.9): 30 resamples with replacement of the cached training images per class at the census budget; ImageNet with 20 null replicates per resample, CIFAR-100 and DTD with 200.

| model    | dataset   | images/class | null rep. | excess (reference) | bootstrap mean | bootstrap s.d. | fraction negative |
|----------|-----------|--------------|-----------|--------------------|----------------|----------------|-------------------|
| ViT-T    | ImageNet  | 100          | 20        | -0.0055            | -0.0051        | 0.0011         | 1.00              |
| ViT-S    | ImageNet  | 100          | 20        | -0.0126            | -0.0122        | 0.0007         | 1.00              |
| ViT-B    | ImageNet  | 100          | 20        | -0.0165            | -0.0160        | 0.0006         | 1.00              |
| ViT-L    | ImageNet  | 100          | 20        | -0.0160            | -0.0160        | 0.0004         | 1.00              |
| DINO-B   | ImageNet  | 100          | 20        | -0.0209            | -0.0208        | 0.0003         | 1.00              |
| DINOv2-S | ImageNet  | 100          | 20        | -0.0164            | -0.0162        | 0.0004         | 1.00              |
| DINOv2-B | ImageNet  | 100          | 20        | -0.0167            | -0.0164        | 0.0002         | 1.00              |
| DINOv2-L | ImageNet  | 100          | 20        | -0.0136            | -0.0134        | 0.0001         | 1.00              |
| DINOv2-G | ImageNet  | 100          | 20        | -0.0080            | -0.0083        | 0.0004         | 1.00              |
| CLIP-B   | ImageNet  | 100          | 20        | -0.0068            | -0.0069        | 0.0009         | 1.00              |
| CLIP-L   | ImageNet  | 100          | 20        | -0.0094            | -0.0095        | 0.0006         | 1.00              |
| SigLIP-B | ImageNet  | 100          | 20        | -0.0080            | -0.0081        | 0.0006         | 1.00              |
| ViT-T    | CIFAR-100 | 500          | 200       | -0.0064            | -0.0059        | 0.0018         | 1.00              |
| ViT-S    | CIFAR-100 | 500          | 200       | -0.0003            | -0.0004        | 0.0012         | 0.67              |
| ViT-B    | CIFAR-100 | 500          | 200       | +0.0057            | +0.0055        | 0.0011         | 0.00              |
| ViT-L    | CIFAR-100 | 500          | 200       | -0.0205            | -0.0205        | 0.0006         | 1.00              |
| DINO-B   | CIFAR-100 | 500          | 200       | -0.0316            | -0.0315        | 0.0006         | 1.00              |
| DINOv2-S | CIFAR-100 | 500          | 200       | -0.0306            | -0.0306        | 0.0004         | 1.00              |
| DINOv2-B | CIFAR-100 | 500          | 200       | -0.0313            | -0.0313        | 0.0004         | 1.00              |
| DINOv2-L | CIFAR-100 | 500          | 200       | -0.0348            | -0.0348        | 0.0003         | 1.00              |
| DINOv2-G | CIFAR-100 | 500          | 200       | -0.0395            | -0.0395        | 0.0002         | 1.00              |
| CLIP-B   | CIFAR-100 | 500          | 200       | -0.0195            | -0.0192        | 0.0007         | 1.00              |
| CLIP-L   | CIFAR-100 | 500          | 200       | -0.0181            | -0.0183        | 0.0009         | 1.00              |
| SigLIP-B | CIFAR-100 | 500          | 200       | -0.0220            | -0.0218        | 0.0008         | 1.00              |
| ViT-T    | DTD       | 80           | 200       | -0.0142            | -0.0141        | 0.0016         | 1.00              |
| ViT-S    | DTD       | 80           | 200       | -0.0186            | -0.0189        | 0.0020         | 1.00              |
| ViT-B    | DTD       | 80           | 200       | -0.0208            | -0.0212        | 0.0017         | 1.00              |
| ViT-L    | DTD       | 80           | 200       | -0.0311            | -0.0302        | 0.0017         | 1.00              |
| DINO-B   | DTD       | 80           | 200       | -0.0272            | -0.0267        | 0.0024         | 1.00              |
| DINOv2-S | DTD       | 80           | 200       | -0.0179            | -0.0171        | 0.0025         | 1.00              |
| DINOv2-B | DTD       | 80           | 200       | -0.0350            | -0.0345        | 0.0016         | 1.00              |
| DINOv2-L | DTD       | 80           | 200       | -0.0420            | -0.0417        | 0.0016         | 1.00              |
| DINOv2-G | DTD       | 80           | 200       | -0.0386            | -0.0387        | 0.0026         | 1.00              |
| CLIP-B   | DTD       | 80           | 200       | -0.0240            | -0.0242        | 0.0021         | 1.00              |
| CLIP-L   | DTD       | 80           | 200       | -0.0266            | -0.0270        | 0.0022         | 1.00              |
| SigLIP-B | DTD       | 80           | 200       | -0.0233            | -0.0234        | 0.0014         | 1.00              |

Table 3: (continued)

(c) Joint sensitivity of the genuine count:  $z_{\text{joint}}$  against null, bootstrap and estimator noise, and (in parentheses) the number of the 30 centroid resamples in which the cell is genuine.

| model    | IN                     | C100                  | C10                    | DTD                    | FMNIST                  | MNIST                 |
|----------|------------------------|-----------------------|------------------------|------------------------|-------------------------|-----------------------|
| ViT-T    | -1.1 (0) <sup>o</sup>  | -0.7 (0) <sup>o</sup> | -0.6 (0) <sup>o</sup>  | -1.9 (12) <sup>†</sup> | +1.3 (0) <sup>o</sup>   | -0.9 (0) <sup>o</sup> |
| ViT-S    | -3.6 (30)              | -0.0 (0) <sup>o</sup> | -2.4 (29)              | -3.0 (30)              | -1.5 (30) <sup>o‡</sup> | -1.3 (0) <sup>o</sup> |
| ViT-B    | -5.2 (30)              | +0.8 (0) <sup>o</sup> | -2.1 (1) <sup>†</sup>  | -3.9 (30)              | -1.9 (30) <sup>†</sup>  | -0.3 (0) <sup>o</sup> |
| ViT-L    | -5.0 (30)              | -4.0 (30)             | -2.4 (30)              | -6.4 (30)              | -2.2 (30)               | -1.4 (0) <sup>o</sup> |
| DINO-B   | -8.9 (30)              | -7.6 (30)             | -2.4 (30)              | -4.4 (30)              | -0.1 (0) <sup>o</sup>   | -1.4 (0) <sup>o</sup> |
| DINOv2-S | -15.9 (30)             | -9.7 (30)             | -2.2 (30)              | -3.1 (30)              | -1.3 (0) <sup>o</sup>   | +0.2 (0) <sup>o</sup> |
| DINOv2-B | -31.7 (30)             | -14.7 (30)            | -4.9 (30)              | -8.0 (30)              | -2.6 (30)               | -0.5 (0) <sup>o</sup> |
| DINOv2-L | -30.7 (30)             | -21.6 (30)            | -6.0 (30)              | -10.6 (30)             | -1.3 (30) <sup>o‡</sup> | -0.4 (0) <sup>o</sup> |
| DINOv2-G | -11.5 (30)             | -28.2 (30)            | -6.5 (30)              | -8.2 (30)              | -1.8 (23) <sup>†</sup>  | -0.7 (0) <sup>o</sup> |
| CLIP-B   | -1.3 (0) <sup>o</sup>  | -3.1 (30)             | -1.3 (0) <sup>o</sup>  | -3.5 (30)              | -1.4 (0) <sup>o</sup>   | +0.3 (0) <sup>o</sup> |
| CLIP-L   | -2.1 (11) <sup>†</sup> | -3.5 (30)             | -1.3 (0) <sup>o</sup>  | -4.2 (30)              | -1.5 (0) <sup>o</sup>   | -0.0 (0) <sup>o</sup> |
| SigLIP-B | -1.7 (0) <sup>o</sup>  | -4.3 (30)             | -1.7 (29) <sup>†</sup> | -4.0 (30)              | -1.8 (30) <sup>†</sup>  | -1.3 (0) <sup>o</sup> |

Table 3: (continued) (c) Per cell:  $z_{\text{joint}} = \text{excess} / \sqrt{\sigma_{\text{null}}^2 + \sigma_{\text{boot}}^2 + \sigma_{\text{est}}^2}$  and the number of resamples in which the cell is genuine under BH over the 72 cells; <sup>o</sup>: not genuine in the record; <sup>†</sup>: record-genuine but failing one of the two criteria ( $z_{\text{joint}} \leq -2$ , genuine in  $\geq 27$  of 30); <sup>‡</sup>: not record-genuine but passing one. Counts: record 44/72 (18/24);  $z_{\text{joint}} \leq -2$ : 39/72 (18/24); genuine in  $\geq 27$  of 30 resamples: 42/72 (17/24). (d) Class-count control under the record (means over subset seeds): the excess shrinks with the number of classes for coherent and random subsets alike, so magnitudes are not comparable across class counts and the verdict is what carries across datasets; “below” = subset seeds whose real value lies below the null at uncorrected  $p \leq 0.05$ ; NC adv = prototype-classifier advantage of the Poincaré readout in pp. (e) Raw  $\delta$  within architecture: fine-tuning on a task without class hierarchy raises it and it falls across transformer depth.

(d) Class count: ImageNet subsets of  $C$  classes, random (spanning the hierarchy) or WordNet-coherent (siblings, effectively flat), means over the subset seeds. The excess shrinks with the number of classes for coherent and random subsets alike, so magnitudes are not comparable across class counts; the verdict is what carries across datasets.

| model    | $C$  | random subsets (span the hierarchy) |        |       |        | WordNet-coherent (siblings) |        |       |        |
|----------|------|-------------------------------------|--------|-------|--------|-----------------------------|--------|-------|--------|
|          |      | $\hat{\delta}_{99.9}$               | excess | below | NC adv | $\hat{\delta}_{99.9}$       | excess | below | NC adv |
| DINOv2-L | 10   | 0.030                               | -0.134 | 5/5   | +1.04  | 0.068                       | -0.098 | 5/5   | +0.08  |
| DINOv2-L | 20   | 0.034                               | -0.091 | 5/5   | +1.32  | 0.071                       | -0.065 | 5/5   | +0.12  |
| DINOv2-L | 50   | 0.032                               | -0.046 | 5/5   | +1.58  | 0.050                       | -0.042 | 5/5   | +0.53  |
| DINOv2-L | 100  | 0.031                               | -0.032 | 5/5   | +1.46  | 0.044                       | -0.031 | 5/5   | +0.77  |
| DINOv2-L | 200  | 0.031                               | -0.022 | 5/5   | +1.45  | 0.041                       | -0.023 | 5/5   | +0.59  |
| DINOv2-L | 500  | 0.030                               | -0.016 | 2/2   | +1.01  | 0.031                       | -0.018 | 2/2   | +0.82  |
| DINOv2-L | 1000 | 0.030                               | -0.014 | 1/1   | +0.57  | —                           | —      | —     | —      |

Table 3: (continued)

(e) Training moves the raw reading within an architecture: one row per intervention. Statistic: the raw supremum  $\delta_{\text{norm}}$ , the largest four-point defect over sampled quadruples divided by the diameter, on 512 sampled points; fine-tuning row: 1000 CIFAR-100 test images before and after a non-hierarchical (colour) fine-tuning of DINOv2-S,  $5 \times 10^4$  quadruples; depth row: DINOv2-B hidden states per layer.

| intervention                            | model    | $\delta$ before | $\delta$ after | change |
|-----------------------------------------|----------|-----------------|----------------|--------|
| non-hierarchical fine-tuning            | DINOv2-S | 0.095           | 0.161          | +69%   |
| across depth (layer 1 $\rightarrow$ 12) | DINOv2-B | 0.147           | 0.058          | -61%   |

Table 3: (continued)

## B.2 IS A LOW RAW READING EVIDENCE? CALIBRATION ON REFERENCE GEOMETRIES AND SYNTHETIC CLUSTERS

(a) Reference geometries.

| space                                                   | absolute $\delta$   | $\delta_{\text{norm}}$              |
|---------------------------------------------------------|---------------------|-------------------------------------|
| balanced binary tree (depth 10)                         | 0.000               | 0.000                               |
| $\mathbb{H}^2$ ( $K = -1$ ), region radius $R=2/4/8/16$ | 0.65/0.69/0.69/0.69 | 0.162/0.087/0.043/0.022             |
| uniform $S^{99}$ (chord / geodesic)                     | 0.25/0.37           | 0.143/0.179                         |
| iid Gaussian ( $n=1000$ ), $d=192, \dots, 1536$         | —                   | 0.104/0.081/0.074/0.061/0.055/0.046 |

Table 4: **A low raw  $\delta$  is not evidence of hierarchy, and the spectrum excess certifies structure beyond the second moments, not depth.** (a) Calibration on reference geometries: the estimator recovers the tree zero and the four-point constant of  $\mathbb{H}^2$ , spheres stay high, and iid Gaussians drift low as  $d$  grows, the dimension confound; the ordering trees  $<$  hyperbolic  $<$  spherical holds for hyperbolic regions of radius  $R \geq 4$ . Gaussian row: the sampled supremum from the released control file, the statistic the curvature rule uses; tree/ $\mathbb{H}^2$ /sphere rows re-verified synthetically at every freeze. (b) Excess of synthetic clouds (means over seeds; within/between noise ratio in parentheses) under the spectrum null A, with Gaussian coefficients and with Haar-rotated coefficients (exact sample spectrum), and under the hub-randomizing null B, which keeps every cluster and resamples the hub configuration. A pure star sits far below null A under either construction, so null A certifies structure beyond the second moments, not depth; null B also reports the star as hierarchical, since a near-regular simplex of hubs already minimizes  $\delta$ , which is why the depth test of Table 9 is read against a matched star rather than against null B. Under the Haar construction the two-level hierarchy sits  $2\text{--}3\times$  below the star; under the plain Gaussian null the mid-radius star scores  $-0.104$  and the  $6\times 5$  hierarchy  $-0.14$ , a fraction of either’s excess.

(b) Synthetic clusters: what each null certifies.

| synthetic cloud ( $n=1000$ , $d=768$ ) | $\hat{\delta}$ | spectrum null A |        | hub-randomizing null B |        |
|----------------------------------------|----------------|-----------------|--------|------------------------|--------|
|                                        |                | Gaussian        | Haar   | Gaussian               | Haar   |
| 30-cluster star, tight (0.1)           | 0.055          | -0.115          | -0.115 | -0.126                 | -0.058 |
| 30-cluster star, mid (0.3)             | 0.059          | -0.104          | -0.103 | -0.114                 | -0.049 |
| 30-cluster star, loose (0.6)           | 0.061          | -0.084          | -0.082 | -0.090                 | -0.033 |
| 2-level hierarchy $6 \times 5$ (0.3)   | 0.044          | -0.143          | -0.143 | -0.113                 | -0.112 |

Table 4: (continued)

### B.3 IS THE STRUCTURE BEYOND THE SECOND MOMENTS GENUINE? THE VISION CENSUS UNDER EVERY CONSTRUCTION

**What each null preserves.** The spectrum-matched null, Gaussian or Haar, preserves  $n$ ,  $d$ , the mean and the covariance spectrum and destroys all higher-order structure; the PC-permutation null preserves each principal component’s marginal and destroys the dependence across components; the label shuffle keeps the point cloud itself and destroys the embedding–label correspondence; and the hub-randomizing null keeps each cluster’s geometry and hub assignment and destroys the hub arrangement, which is why it serves the depth test only when read against a matched star. “Genuine” in the text always means beyond the first of these. In plain terms, the spectrum null is a cloud with the same shape as the real one and no structure inside it; the hub null of Section 5 keeps every cluster of the real cloud and only rearranges where the clusters sit.

(a) The census of record per cell: Haar spectrum-matched null with  $Q$  orthogonal to the all-ones vector, 99.9th-percentile statistic, 200 replicates.

| model    | IN      |     |      | C100    |     |      | C10     |     |      | DTD    |     |      | FMNIST  |     |      | MNIST   |     |      |
|----------|---------|-----|------|---------|-----|------|---------|-----|------|--------|-----|------|---------|-----|------|---------|-----|------|
|          | exc.    | $r$ | $p$  | exc.    | $r$ | $p$  | exc.    | $r$ | $p$  | exc.   | $r$ | $p$  | exc.    | $r$ | $p$  | exc.    | $r$ | $p$  |
| ViT-T    | −0.005° | 169 | .159 | −0.006° | 151 | .249 | −0.012° | 148 | .264 | −0.014 | 195 | .030 | +0.024° | 19  | .905 | −0.022° | 168 | .164 |
| ViT-S    | −0.012  | 200 | .005 | −0.000° | 112 | .443 | −0.042  | 199 | .010 | −0.018 | 199 | .010 | −0.027° | 192 | .045 | −0.034° | 183 | .090 |
| ViT-B    | −0.016  | 200 | .005 | +0.006° | 42  | .791 | −0.038  | 196 | .025 | −0.020 | 200 | .005 | −0.035  | 198 | .015 | −0.007° | 122 | .393 |
| ViT-L    | −0.016  | 200 | .005 | −0.020  | 200 | .005 | −0.037  | 200 | .005 | −0.030 | 200 | .005 | −0.041  | 200 | .005 | −0.031° | 187 | .070 |
| DINO-B   | −0.020  | 200 | .005 | −0.031  | 200 | .005 | −0.042  | 199 | .010 | −0.026 | 200 | .005 | −0.002° | 104 | .483 | −0.030° | 182 | .095 |
| DINOv2-S | −0.016  | 200 | .005 | −0.030  | 200 | .005 | −0.033  | 200 | .005 | −0.017 | 200 | .005 | −0.025° | 184 | .085 | +0.003° | 79  | .607 |
| DINOv2-B | −0.017  | 200 | .005 | −0.031  | 200 | .005 | −0.065  | 200 | .005 | −0.034 | 200 | .005 | −0.056  | 199 | .010 | −0.012° | 142 | .294 |
| DINOv2-L | −0.014  | 200 | .005 | −0.034  | 200 | .005 | −0.084  | 200 | .005 | −0.041 | 200 | .005 | −0.024° | 186 | .075 | −0.009° | 135 | .328 |
| DINOv2-G | −0.008  | 200 | .005 | −0.039  | 200 | .005 | −0.077  | 200 | .005 | −0.037 | 200 | .005 | −0.035  | 197 | .020 | −0.018° | 163 | .189 |
| CLIP-B   | −0.007° | 172 | .144 | −0.019  | 200 | .005 | −0.022° | 184 | .085 | −0.023 | 200 | .005 | −0.032° | 185 | .080 | +0.005° | 81  | .597 |
| CLIP-L   | −0.009  | 197 | .020 | −0.018  | 200 | .005 | −0.020° | 180 | .104 | −0.026 | 200 | .005 | −0.033° | 186 | .075 | −0.001° | 97  | .517 |
| SigLIP-B | −0.008° | 189 | .060 | −0.022  | 200 | .005 | −0.027  | 197 | .020 | −0.023 | 200 | .005 | −0.033  | 195 | .030 | −0.022° | 176 | .124 |

Table 5: **Structure beyond the second moments is the rule at the class level under every construction of the null and statistic, and the verdict does not depend on the metric.** (a) Per cell: excess of  $\hat{\delta}_{99.9}$  over the null mean,  $r$  = replicates above the real value,  $p = (1 + \#\{\text{null} \leq \text{real}\})/201$ ; genuine = Benjamini–Hochberg-corrected  $p \leq 0.05$  over the 72 cells (°: not genuine). Genuine cells: 44/72 overall, 18/24 on ImageNet+CIFAR-100; sign agreement with the original 5-replicate Gaussian×supremum census on the 72 comparable cells: 65/72. (b) Five symbols per cell in the order centered Haar×p99.9 (the record,  $Q \perp 1$ ), uncentered Haar×p99.9, Haar×supremum, Gaussian×p99.9, Gaussian×supremum, • = genuine under that construction; genuine counts 44/72, 49/72, 46/72, 52/72, 52/72. The percentage is the record excess divided by the mean null reading (ImageNet -33% to -5%; all cells -75% to +18%); the last column is the ImageNet excess of the supremum statistic under the same Haar null with its rank. The centered Haar construction reproduces the centered spectrum exactly, the uncentered one differs by a term of order  $1/\sqrt{n}$  that moves the verdict only on the ten-class datasets; the Gaussian one does not at small  $n$  and inflates excesses for low-rank clouds (Table 3). The supremum is decided by a few extreme quadruples and disagrees with the 99.9th percentile in both directions on DINOv2-S/B/G ImageNet.

(b) Verdict under the five null×statistic constructions and the excess as a fraction of the null reading.

| model    | IN        | C100      | C10       | DTD       | FMNIST    | MNIST     | sup. IN exc. ( $r$ ) |
|----------|-----------|-----------|-----------|-----------|-----------|-----------|----------------------|
| ViT-T    | ○○●● −5%  | ○○●● −5%  | ○○○○ −9%  | ●●●● −11% | ○○○○ +18% | ○○○○ −14% | −0.020 (199)         |
| ViT-S    | ●●●● −16% | ○○●● −0%  | ●●●● −33% | ●●●● −16% | ●●●● −21% | ○○○○ −21% | −0.020 (200)         |
| ViT-B    | ●●●● −23% | ○○○○ +5%  | ●●●● −30% | ●●●● −18% | ●●●● −26% | ○○○○ −5%  | −0.022 (200)         |
| ViT-L    | ●●●● −22% | ●●●● −18% | ●●●● −31% | ●●●● −28% | ●●●● −31% | ○○●● −20% | −0.027 (200)         |
| DINO-B   | ●●●● −27% | ●●●● −28% | ●●●● −32% | ●●●● −22% | ○○○○ −1%  | ○○○○ −21% | +0.000 (96)          |
| DINOv2-S | ●●●● −23% | ●●●● −31% | ●●●● −28% | ●●●● −14% | ○○○○ −19% | ○○○○ +2%  | +0.017 (0)           |
| DINOv2-B | ●●●● −33% | ●●●● −38% | ●●●● −59% | ●●●● −31% | ●●●● −40% | ○○○○ −8%  | +0.018 (0)           |
| DINOv2-L | ●●●● −31% | ●●●● −48% | ●●●● −75% | ●●●● −37% | ●●●● −19% | ○○○○ −6%  | +0.006 (0)           |
| DINOv2-G | ●●●● −20% | ●●●● −56% | ●●●● −75% | ●●●● −34% | ●●●● −26% | ○○○○ −11% | +0.020 (0)           |
| CLIP-B   | ○○●● −7%  | ●●●● −16% | ○○●● −17% | ●●●● −19% | ○○○○ −19% | ○○○○ +4%  | −0.023 (199)         |
| CLIP-L   | ●●●● −11% | ●●●● −16% | ○○●● −17% | ●●●● −21% | ○○○○ −22% | ○○○○ −0%  | −0.016 (198)         |
| SigLIP-B | ○○●● −9%  | ●●●● −18% | ●●●● −23% | ●●●● −19% | ●●●● −23% | ○○●● −16% | −0.015 (194)         |

Table 5: (continued)

(c) The same census on cosine geometry (L2-normalized centroids, geodesic distances, Haar null on the normalized cloud): robustness, not the record.

| model    | IN     |     | C100                |     | C10    |     | DTD    |     | FMNIST              |     | MNIST               |     |
|----------|--------|-----|---------------------|-----|--------|-----|--------|-----|---------------------|-----|---------------------|-----|
|          | exc.   | $r$ | exc.                | $r$ | exc.   | $r$ | exc.   | $r$ | exc.                | $r$ | exc.                | $r$ |
| ViT-T    | -0.012 | 200 | -0.014 <sup>o</sup> | 191 | -0.043 | 198 | -0.017 | 199 | +0.007 <sup>o</sup> | 73  | -0.026 <sup>o</sup> | 167 |
| ViT-S    | -0.017 | 200 | -0.017              | 200 | -0.073 | 200 | -0.021 | 200 | -0.049              | 200 | -0.036 <sup>o</sup> | 183 |
| ViT-B    | -0.022 | 200 | -0.006 <sup>o</sup> | 170 | -0.066 | 200 | -0.021 | 200 | -0.041              | 199 | -0.010 <sup>o</sup> | 130 |
| ViT-L    | -0.021 | 200 | -0.024              | 200 | -0.080 | 200 | -0.024 | 200 | -0.066              | 200 | -0.043              | 196 |
| DINO-B   | -0.021 | 200 | -0.035              | 200 | -0.070 | 200 | -0.024 | 200 | -0.006 <sup>o</sup> | 116 | -0.038 <sup>o</sup> | 193 |
| DINOv2-S | -0.014 | 200 | -0.033              | 200 | -0.084 | 200 | -0.018 | 200 | -0.054              | 200 | -0.004 <sup>o</sup> | 113 |
| DINOv2-B | -0.013 | 200 | -0.032              | 200 | -0.140 | 200 | -0.021 | 200 | -0.081              | 200 | -0.019 <sup>o</sup> | 147 |
| DINOv2-L | -0.010 | 200 | -0.029              | 200 | -0.151 | 200 | -0.024 | 200 | -0.051              | 200 | -0.020 <sup>o</sup> | 145 |
| DINOv2-G | -0.006 | 200 | -0.029              | 200 | -0.169 | 200 | -0.017 | 200 | -0.064              | 200 | -0.023 <sup>o</sup> | 157 |
| CLIP-B   | -0.017 | 200 | -0.023              | 200 | -0.039 | 197 | -0.031 | 200 | -0.038 <sup>o</sup> | 185 | +0.002 <sup>o</sup> | 93  |
| CLIP-L   | -0.017 | 200 | -0.025              | 200 | -0.045 | 199 | -0.031 | 200 | -0.038 <sup>o</sup> | 192 | -0.008 <sup>o</sup> | 124 |
| SigLIP-B | -0.013 | 200 | -0.025              | 200 | -0.053 | 200 | -0.027 | 200 | -0.052              | 199 | -0.044              | 197 |

Table 5: (continued) (c) Cosine reading, BH over 72 cells: sign-negative 70/72, genuine 56/72 and 22/24 on ImageNet+CIFAR-100, verdict agreement with the Euclidean record 60/72 cells. Text under the same cosine protocol (padding-free embeddings): genuine 8/15: gpt2-m, gpt2-l, gpt2-xl, pythia-410m, pythia-1b, pythia-2b8, olmo-1b, gte-qwen2. The angular tree: sibling-triplet agreement on CIFAR-100 for DINOv2-L is 0.91 under cosine against 0.71 under Euclidean distance. (d) Two self-supervised ResNet-50 backbones read the same clustered structure as the ViTs.

(d) Backbones outside the ViT panel on the ImageNet centroid store (spectrum null, 20 replicates).

| model                    | $d$  | $\hat{\delta}$ | excess ( $z$ ) | reps above |
|--------------------------|------|----------------|----------------|------------|
| Barlow Twins (ResNet-50) | 2048 | 0.075          | -0.039 (-4.8)  | 20/20      |
| BYOL (ResNet-50)         | 2048 | 0.073          | -0.020 (-2.4)  | 20/20      |

Table 5: (continued)

#### B.4 THE MODEL PANEL AND EXTRACTION

| type                    | model                              | params       |
|-------------------------|------------------------------------|--------------|
| Supervised ViT (vision) | ViT-T (i21k)                       | 5M           |
| Supervised ViT (vision) | ViT-S (i21k)                       | 22M          |
| Supervised ViT (vision) | ViT-B (i21k)                       | 86M          |
| Supervised ViT (vision) | ViT-L (i21k)                       | 307M         |
| SSL (vision)            | DINO-B                             | 86M          |
| SSL (vision)            | DINOv2-S                           | 22M          |
| SSL (vision)            | DINOv2-B                           | 86M          |
| SSL (vision)            | DINOv2-L                           | 307M         |
| SSL (vision)            | DINOv2-G                           | 1.1B         |
| Contrastive (vision)    | CLIP-B                             | 86M          |
| Contrastive (vision)    | CLIP-L                             | 307M         |
| Contrastive (vision)    | SigLIP-B                           | 86M          |
| Causal LM               | GPT-2 S                            | 117M         |
| Causal LM               | GPT-2 M                            | 345M         |
| Causal LM               | GPT-2 L                            | 774M         |
| Causal LM               | GPT-2 XL                           | 1.5B         |
| Causal LM               | Pythia-410M                        | 410M         |
| Causal LM               | Pythia-1B                          | 1.0B         |
| Causal LM               | Pythia-2.8B                        | 2.8B         |
| Causal LM               | OLMo-1B                            | 1.0B         |
| Text embedder           | BGE-base                           | 110M         |
| Text embedder           | BGE-large                          | 335M         |
| Text embedder           | GTE-base                           | 110M         |
| Text embedder           | GTE-large                          | 335M         |
| Text embedder           | GTE-Qwen2-1.5B                     | 1.5B         |
| Text embedder           | E5-base                            | 110M         |
| Text embedder           | E5-large                           | 335M         |
| Controls (vision)       | DeiT-B/16 (IN-1k, no distillation) | 86M          |
| Controls (vision)       | ViT-B/16 augreg (IN-1k)            | 86M          |
| Controls (vision)       | MERU ViT-S/B/L and CLIP twins      | 22M/86M/307M |
| Controls (vision)       | Barlow Twins, BYOL (ResNet-50)     | 24M          |

**Table 6: The model panel.** 12 vision backbones (all in the census; the 10 with cached per-dataset features, all but DINO-B and SigLIP-B, form the task grid), 8 causal LMs, 7 text embedders, and the vision models used only as controls. Class centroids average 100 cached training images per class on ImageNet and the full cached split, 80–6,000 images per class, on the other datasets. Feature extraction: vision backbones use the encoder’s default pooled embedding (timm, num\_classes=0, CLS or global pool per architecture); causal LMs use the last-token hidden state, extracted one prompt at a time (padding-free: left-padded batching alters GPT-2’s hidden states, Table 7); text embedders use their recommended pooling (CLS for BGE, mean for GTE/E5, last token for GTE-Qwen2); CLIP/SigLIP text towers use their text encoders.

#### B.5 IS TEXT THE SAME? THE TEXT CENSUS UNDER THE RECORD

**Extraction, anisotropy and templates.** Extraction is part of the measurement: left-padded batching moves GPT-2’s  $\hat{\delta}$  by up to 0.018, of the order of the fragile excesses; the padding-free protocol is the record. Genuineness is not a proxy for anisotropy: GTE-base has the highest effective rank and is not genuine, and the only LLM-backed embedder tested (GTE-Qwen2-1.5B) is not genuine under the record. Causal-LM values are template-sensitive but the scale ordering is template-robust (Table 7).

(a) The census of record on the 1000 ImageNet class prompts: padding-free extraction, Haar null, 99.9th-percentile statistic, 200 replicates.

| model          | type      | $d$  | $\hat{\delta}_{99.9}$ | excess  | record<br>exc./null | $r/200$ | $p$  | $2 \times 2$ | cosine<br>exc. | $r$ |
|----------------|-----------|------|-----------------------|---------|---------------------|---------|------|--------------|----------------|-----|
| GPT-2 S        | causal LM | 768  | 0.077                 | -0.013  | -15%                | 197     | .020 | ●●●○         | +0.000°        | 101 |
| GPT-2 M        | causal LM | 1024 | 0.052                 | -0.005° | -9%                 | 170     | .154 | ○○○○         | -0.019         | 200 |
| GPT-2 L        | causal LM | 1280 | 0.053                 | -0.023  | -30%                | 200     | .005 | ●●●●         | -0.016         | 200 |
| GPT-2 XL       | causal LM | 1600 | 0.048                 | -0.027  | -36%                | 200     | .005 | ●●●●         | -0.018         | 200 |
| Pythia-410M    | causal LM | 1024 | 0.067                 | -0.018  | -21%                | 200     | .005 | ●●●●         | -0.024         | 200 |
| Pythia-1B      | causal LM | 2048 | 0.066                 | -0.015  | -18%                | 199     | .010 | ●●●●         | -0.026         | 200 |
| Pythia-2.8B    | causal LM | 2560 | 0.066                 | -0.017  | -20%                | 200     | .005 | ●●●●         | -0.023         | 200 |
| OLMo-1B        | causal LM | 2048 | 0.051                 | -0.024  | -32%                | 200     | .005 | ●●●●         | -0.032         | 200 |
| BGE-base       | embedder  | 768  | 0.078                 | +0.003° | +4%                 | 26      | .871 | ○○○○         | -0.002°        | 151 |
| BGE-large      | embedder  | 1024 | 0.078                 | +0.002° | +3%                 | 49      | .756 | ○○○○         | -0.003°        | 170 |
| GTE-base       | embedder  | 768  | 0.076                 | +0.004° | +6%                 | 11      | .945 | ○○○○         | +0.001°        | 79  |
| GTE-large      | embedder  | 1024 | 0.075                 | +0.003° | +3%                 | 36      | .821 | ○○○○         | -0.001°        | 131 |
| GTE-Qwen2-1.5B | embedder  | 1536 | 0.078                 | +0.002° | +3%                 | 58      | .711 | ○○○○         | -0.006         | 199 |
| E5-base        | embedder  | 768  | 0.074                 | +0.004° | +6%                 | 15      | .925 | ○○○○         | +0.000°        | 97  |
| E5-large       | embedder  | 1024 | 0.071                 | +0.002° | +3%                 | 44      | .781 | ○○○○         | -0.001°        | 119 |

Table 7: **In text, structure beyond the second moments depends on recipe and scale: genuine at every Pythia scale and at the larger GPT-2 sizes, at the margin for GPT-2 S, absent for GPT-2 M and for the sentence embedders on this probe.** (a)  $r$  = replicates above the real value,  $p$  the left-tail add-one  $p$ -value; genuine = BH-corrected  $p \leq 0.05$  over the 15 models (°: not genuine); exc./null: excess as a fraction of the null reading; the  $2 \times 2$  column gives the verdicts under Haar $\times$ p99.9, Haar $\times$ supremum, Gaussian $\times$ p99.9, Gaussian $\times$ supremum (● genuine); cosine: the same census on cosine geometry. Genuine 7/15; GPT-2 S is genuine at  $p = .020$  with excess -0.013 and GPT-2 M is not ( $p = .154$ , -0.005), while L and XL are genuine under every construction. GTE-base has the highest effective rank and is not genuine, so genuineness is not a proxy for anisotropy; the only LLM-backed embedder (GTE-Qwen2-1.5B) is not genuine. (b) Causal-LM values are template-sensitive (range of raw  $\hat{\delta}$  up to 0.065 for causal LMs against  $\leq 0.012$  for embedders; “gloss only” contains no class name) but the scale ordering is template-robust: with only 3 null replicates per cell the starred S and M cells are borderline, as the 200-replicate record in (a) shows, whereas the L/XL cells are not in doubt. (c) Precision is irrelevant (fp16 equals fp32 to four decimals); for GPT-2, left-padded batching changes the hidden states (mean cosine to the padding-free extraction 0.98) and moves  $\hat{\delta}$  by up to 0.018, while OLMo handles left padding correctly and is invariant, which is why padding-free extraction is the record.

(b) Prompt templates: raw  $\hat{\delta}$  per template and, for the four GPT-2 sizes, the template-conditioned excess over the spectrum null (3 replicates; \*:  $|z| \geq 2$ ) and its fraction of the null.

| template                  | raw $\hat{\delta}$ |       |         |         | GPT-2 excess (padding-free) |         |         |          | GPT-2 excess / null |      |      |      |
|---------------------------|--------------------|-------|---------|---------|-----------------------------|---------|---------|----------|---------------------|------|------|------|
|                           | BGE-b              | E5-b  | GPT-2 S | GPT-2 M | S                           | M       | L       | XL       | S                   | M    | L    | XL   |
| photo                     | 0.124              | 0.114 | 0.131   | 0.091   | -0.016*                     | -0.019* | -0.043* | -0.050*  | -11%                | -19% | -34% | -41% |
| name_only                 | 0.118              | 0.119 | 0.145   | 0.122   | -0.006                      | -0.007  | -0.048* | -0.051*  | -6%                 | -5%  | -39% | -43% |
| image                     | 0.125              | 0.117 | 0.110   | 0.110   | -0.018*                     | -0.028* | -0.050* | -0.049*  | -13%                | -26% | -41% | -41% |
| closeup                   | 0.126              | 0.120 | 0.086   | 0.100   | +0.007                      | -0.018* | -0.044* | -0.049*  | +5%                 | -18% | -36% | -40% |
| this_is                   | 0.125              | 0.115 | 0.080   | 0.134   | -0.032*                     | -0.022* | -0.042* | -0.046*  | -19%                | -22% | -34% | -38% |
| wild                      | 0.115              | 0.116 | 0.089   | 0.104   | -0.023*                     | -0.067* | -0.014  | -0.010*  | -16%                | -43% | -11% | -8%  |
| def_pair                  | 0.119              | 0.108 | 0.105   | 0.081   | -0.002                      | -0.019* | -0.036* | -0.035*  | -1%                 | -21% | -28% | -29% |
| gloss_only                | 0.115              | 0.115 | 0.124   | 0.107   | -0.020*                     | -0.030* | -0.033* | -0.036*  | -14%                | -31% | -27% | -29% |
| concept                   | 0.119              | 0.120 | 0.112   | 0.112   | -0.002                      | -0.003  | -0.043* | -0.041*  | -1%                 | -3%  | -38% | -38% |
| discussion                | 0.121              | 0.116 | 0.105   | 0.098   | -0.010                      | -0.013* | -0.041* | -0.040*  | -7%                 | -17% | -35% | -35% |
| mean over templates       |                    |       |         |         | -0.012                      | -0.023  | -0.040  | -0.041   |                     |      |      |      |
| templates at $ z  \geq 2$ |                    |       |         |         | 5 of 10                     | 8 of 10 | 9 of 10 | 10 of 10 |                     |      |      |      |

Table 7: (continued)

(c) Extraction: the same 1000 prompts, only batch size and precision vary.

| batch size     | GPT-2 M             |                     | OLMo-1B             |                     |
|----------------|---------------------|---------------------|---------------------|---------------------|
|                | fp32 $\hat{\delta}$ | fp16 $\hat{\delta}$ | fp32 $\hat{\delta}$ | fp16 $\hat{\delta}$ |
| 1 (no padding) | 0.0816              | 0.0817              | 0.0817              | 0.0817              |
| 16             | 0.0996              | 0.0997              | 0.0817              | 0.0816              |
| 32             | 0.0905              | 0.0905              | 0.0817              | 0.0817              |

Table 7: (continued)

## B.6 DOES THE PREMISE SURVIVE AT THE SAMPLE LEVEL? THE CENSUS ON PER-IMAGE FEATURES

(a) The census cells at the sample level.

| model    | CIFAR-100 (10 images/class, $n=1000$ ) |                       |                 |           |                 | DTD (22 images/class, $n=1034$ ) |                       |                 |           |                 |
|----------|----------------------------------------|-----------------------|-----------------|-----------|-----------------|----------------------------------|-----------------------|-----------------|-----------|-----------------|
|          | sup. $\hat{\delta}$                    | $\hat{\delta}_{99.9}$ | excess          | exc./null | $r/200$ ( $p$ ) | sup. $\hat{\delta}$              | $\hat{\delta}_{99.9}$ | excess          | exc./null | $r/200$ ( $p$ ) |
| ViT-T    | 0.130                                  | 0.093                 | $-0.0016^\circ$ | -2%       | 134 (0.333)     | 0.124                            | 0.087                 | $-0.0004^\circ$ | -0%       | 111 (0.448)     |
| ViT-S    | 0.121                                  | 0.086                 | $+0.0051^\circ$ | +6%       | 21 (0.896)      | 0.103                            | 0.070                 | $-0.0062^\circ$ | -8%       | 197 (0.020)     |
| ViT-B    | 0.121                                  | 0.082                 | $+0.0031^\circ$ | +4%       | 50 (0.751)      | 0.092                            | 0.060                 | $-0.0080^\circ$ | -12%      | 200 (0.005)     |
| ViT-L    | 0.100                                  | 0.067                 | $-0.0041^\circ$ | -6%       | 185 (0.080)     | 0.092                            | 0.056                 | $-0.0073^\circ$ | -12%      | 200 (0.005)     |
| DINO-B   | 0.104                                  | 0.070                 | $-0.0042^\circ$ | -6%       | 194 (0.035)     | 0.110                            | 0.075                 | $+0.0008^\circ$ | +1%       | 80 (0.602)      |
| DINOv2-S | 0.100                                  | 0.065                 | $-0.0049^\circ$ | -7%       | 200 (0.005)     | 0.145                            | 0.090                 | $+0.0058^\circ$ | +7%       | 0 (1.000)       |
| DINOv2-B | 0.089                                  | 0.050                 | $-0.0090^\circ$ | -15%      | 200 (0.005)     | 0.139                            | 0.073                 | $-0.0019^\circ$ | -3%       | 162 (0.194)     |
| DINOv2-L | 0.087                                  | 0.043                 | $-0.0131^\circ$ | -24%      | 200 (0.005)     | 0.131                            | 0.064                 | $-0.0060^\circ$ | -9%       | 200 (0.005)     |
| DINOv2-G | 0.102                                  | 0.036                 | $-0.0232^\circ$ | -39%      | 200 (0.005)     | 0.136                            | 0.065                 | $-0.0079^\circ$ | -11%      | 200 (0.005)     |
| CLIP-B   | 0.117                                  | 0.080                 | $-0.0060^\circ$ | -7%       | 184 (0.085)     | 0.125                            | 0.089                 | $+0.0021^\circ$ | +2%       | 71 (0.647)      |
| CLIP-L   | 0.132                                  | 0.084                 | $+0.0037^\circ$ | +5%       | 35 (0.826)      | 0.120                            | 0.082                 | $-0.0008^\circ$ | -1%       | 124 (0.383)     |
| SigLIP-B | 0.113                                  | 0.078                 | $-0.0074^\circ$ | -9%       | 199 (0.010)     | 0.114                            | 0.079                 | $-0.0044^\circ$ | -5%       | 184 (0.085)     |

Table 8: **The sample-level reading, the object of prior latent-hyperbolicity claims, sits within null noise in most cells.** The instrument on  $\approx 1000$  stratified training images per cell (per-image features, not centroids): sup.  $\hat{\delta}$  is the raw supremum  $\delta_{\text{norm}}$  the literature reports;  $\hat{\delta}_{99.9}$ , excess, exc./null,  $r/200$  and  $p$  follow Table 2 (Haar spectrum-matched null on the sample cloud, 99.9th-percentile statistic, 200 replicates);  $^\circ$ : not genuine under Benjamini–Hochberg over the 24 cells. The raw supremum lies in the band 0.087–0.145; the excess is not genuine in 14 of 24 cells and genuine in 10 (DINOv2-S, DINOv2-B, DINOv2-L, DINOv2-G, SigLIP-B on CIFAR-100; ViT-S, ViT-B, ViT-L, DINOv2-L, DINOv2-G on DTD), where it reaches at most  $-0.023$ , that is -39% to -7% of the null reading.

(b) A published reading reproduced and calibrated: the setting of Khrulkov et al. (2020), Table 1, ResNet-34 row.

| dataset      | their $\delta_{\text{rel}}$ | our $\delta_{\text{rel}}$ | excess               | rank $r$ | largest $p$ |
|--------------|-----------------------------|---------------------------|----------------------|----------|-------------|
| CIFAR-10     | 0.26                        | $0.275 \pm 0.017$         | $-0.0037 \pm 0.0017$ | 173/200  | 0.333       |
| CIFAR-100    | 0.25                        | $0.261 \pm 0.013$         | $-0.0087 \pm 0.0025$ | 198/200  | 0.030       |
| CUB-200      | 0.25                        | $0.270 \pm 0.012$         | $-0.0043 \pm 0.0020$ | 177/200  | 0.249       |
| MiniImageNet | 0.21                        | $0.203 \pm 0.013$         | $-0.0234 \pm 0.0018$ | 200/200  | 0.005       |

(b) Penultimate features of an ImageNet-pretrained ResNet-34 (torchvision, no substitution) on class-balanced batches of 1500 points; their estimator (exact  $\delta$  on the batch,  $\delta_{\text{rel}} = 2\delta/\text{diam}$ ; mean  $\pm$  s.d. over 10 batches) next to their published value, and on the same clouds the record instrument: excess over the centered Haar null (200 replicates, 99.9th-percentile statistic), the mean rank of the reading among the replicates and the largest left-tail  $p$  over the 10 batches. Every reproduced raw value is within 0.03 of the published one.

Table 8: (continued)

### B.7 IS THE STRUCTURE DEEPER THAN A STAR? THE DEPTH TEST

**Real-data readings by regime.** ImageNet ( $n=1000$ , WordNet  $K=30$ ) lies in the regime where false alarms are controlled: the real cloud is more hierarchical than its anisotropic matched star beyond the noise ( $z \leq -2$ ) in 4 of 12 backbones (ViT-S, ViT-B, ViT-L, DINOv2-L), never less, under Gaussian and under Haar-resampled star hubs alike (Table 9, Figure 4a). CIFAR-100 ( $n=100$ ,  $K=20$ ) lies in the regime where false alarms are not controlled, where pure stars are declared more hierarchical in 17% of runs; its readings (3 of 12 with  $z \leq -2$ , none with  $z \geq +2$ ) are reported without certification (Table 9). Validity domain of the test as run:  $n \approx 1000$ ; at  $n=100$  the false-alarm rate is 0% with  $K=6$  (17 points per cluster) and 17% with  $K \geq 12$ , so the test is not validated below about ten points per cluster. The bar was fixed before the sweep; the restriction to the regime where false alarms are controlled was adopted after it.

**A trained positive control.** Table 9(d) gives the fine-tuned ViT-B/16 controls, cross-entropy alone and with a hierarchical term, and the decoupled comparison that separates them.

(a) The twelve backbones under the stars with Gaussian hubs: depth = excess B of the real centroids minus that of a matched star (negative = more hierarchical above the frame), with  $z$  against the combined spread; the WordNet cut at  $K=10, 30$  and 60 superclasses and the balanced frame.

| model    | C100, $K=20$ |                      | ImageNet, anisotropic star |            |                      |            | bal. frame |
|----------|--------------|----------------------|----------------------------|------------|----------------------|------------|------------|
|          | iso. $z$     | aniso. depth ( $z$ ) | iso. $K=30$ $z$            | $K=10$ $z$ | $K=30$ depth ( $z$ ) | $K=60$ $z$ | real $z$   |
| ViT-T    | +0.02        | -0.025 (-1.43)       | -1.48                      | -1.70      | -0.018 (-1.48)       | -2.53      | -2.08      |
| ViT-S    | +2.34        | -0.008 (-0.51)       | -1.95                      | -1.72      | -0.028 (-2.36)       | -1.78      | -1.72      |
| ViT-B    | +4.47        | +0.003 (+0.23)       | -2.87                      | -2.57      | -0.033 (-3.62)       | -2.23      | -2.82      |
| ViT-L    | +4.20        | -0.017 (-1.49)       | -3.17                      | -2.73      | -0.033 (-4.04)       | -4.69      | -5.57      |
| DINO-B   | +0.86        | -0.038 (-4.24)       | -0.21                      | -0.62      | -0.009 (-0.94)       | -1.11      | -0.98      |
| DINOv2-S | +0.55        | -0.020 (-1.61)       | +0.94                      | +0.26      | +0.008 (+0.66)       | +0.61      | +0.12      |
| DINOv2-B | +5.40        | +0.010 (+0.94)       | -0.93                      | -0.46      | -0.016 (-1.19)       | -0.75      | -0.32      |
| DINOv2-L | +6.77        | +0.013 (+1.20)       | -2.27                      | -1.20      | -0.028 (-2.39)       | -1.25      | -1.70      |
| DINOv2-G | +3.76        | -0.007 (-0.68)       | -1.57                      | -1.00      | -0.029 (-1.78)       | -0.48      | -0.93      |
| CLIP-B   | +1.75        | -0.024 (-2.00)       | -0.72                      | -1.55      | -0.016 (-1.95)       | -2.10      | -1.29      |
| CLIP-L   | +5.68        | -0.013 (-2.34)       | +0.46                      | -1.26      | -0.008 (-0.83)       | -1.40      | -0.68      |
| SigLIP-B | +2.77        | -0.021 (-2.17)       | +0.05                      | -0.76      | -0.011 (-0.90)       | -1.66      | -1.12      |

Table 9: **The depth test certifies in four ImageNet backbones under both matched stars that clusters are oriented toward their hubs, not that the hubs of the frame form a hierarchy, and no hub hierarchy is found in the nine backbones where the decoupled control detects an implanted one, and the test is blind in the other three: none fires once cluster orientations are randomized, whereas a three-level hierarchy implanted at the same noise level is detected and survives that randomization (Table 10); imposing the geometry does not create the structure, and leaf-label supervision can produce it or not.** (a) Isotropic star: Gaussian clouds with each superclass’s RMS spread (the earlier, withdrawn construction); anisotropic star: within each superclass a Haar sample with the cloud’s own covariance, hubs Gaussian at the real hub radius; Haar-resampled hubs: the same clusters with the hubs drawn as a Haar resample of the real hubs. Hub-randomizing Haar null, 10 star seeds; frames: CIFAR-100 coarse labels, ImageNet WordNet cut. The test controls false alarms at  $n=1000$  only (Table 10); CIFAR-100 readings ( $n=100$ ) are reported without certification. Certified on ImageNet ( $n=1000$ , the regime where false alarms are controlled): 4 of 12 backbones (ViT-S, ViT-B, ViT-L, DINOv2-L) with  $z$  from  $-2.36$  to  $-4.04$ ; the withdrawn isotropic star reached  $z=+6.77$  on CIFAR-100. With the hubs of the matched star drawn as a Haar resample of the real hubs (exact hub spectrum, 10 star seeds) the certified set is ViT-S, ViT-B, ViT-L, DINOv2-L, with  $z$  from  $-2.04$  to  $-3.73$ ;  $r_{\text{star}}$  counts the star seeds whose excess B lies at or below the real one, a rank whose resolution is one part in eleven and which does not separate the certified backbones from the others. Balanced frame (complete-linkage WordNet cut with the smallest variance of cluster sizes, chosen before any depth run): certified ViT-T, ViT-B, ViT-L. K sweep: CIFAR-100  $K=5$ : iso 0/12 at  $z \leq -2$ , 7/12 at  $z \geq +2$ ; aniso 0/12 at  $z \leq -2$ , 0/12 at  $z \geq +2$ . CIFAR-100  $K=10$ : iso 0/12 at  $z \leq -2$ , 9/12 at  $z \geq +2$ ; aniso 1/12 at  $z \leq -2$ , 0/12 at  $z \geq +2$ . CIFAR-100  $K=20$ : iso 0/12 at  $z \leq -2$ , 8/12 at  $z \geq +2$ ; aniso 3/12 at  $z \leq -2$ , 0/12 at  $z \geq +2$ . ImageNet  $K=10$ : iso 0/12 at  $z \leq -2$ , 0/12 at  $z \geq +2$ ; aniso 2/12 at  $z \leq -2$ , 0/12 at  $z \geq +2$ . ImageNet  $K=30$ : iso 3/12 at  $z \leq -2$ , 0/12 at  $z \geq +2$ ; aniso 4/12 at  $z \leq -2$ , 0/12 at  $z \geq +2$ . ImageNet  $K=60$ : iso 3/12 at  $z \leq -2$ , 0/12 at  $z \geq +2$ ; aniso 4/12 at  $z \leq -2$ , 0/12 at  $z \geq +2$ . (b) Supervision on leaf labels can produce the depth (the augreg recipe is certified under both stars) and need not (DeiT-B is not detected): the hierarchical label set of IN-21k is not what makes the census ViTs deep; their WordNet alignment is in Table 12.

(a') The same backbones under the star whose hubs are a Haar resample of the real hubs (ImageNet,  $K=30$ ); last column: the decoupling control, the real hubs kept and each cluster's offsets rotated by an independent Haar rotation (mean  $z \pm$  s.d. over 10 seeds; seeds certified at  $z \leq -2$ ).

| model    | star B  | depth ( $z$ )   | $r_{\text{star}}$ | decoupled $z$ (cert.)   |
|----------|---------|-----------------|-------------------|-------------------------|
| ViT-T    | +0.0011 | -0.0184 (-1.45) | 0/10              | -1.69 $\pm$ 0.28 (1/10) |
| ViT-S    | +0.0010 | -0.0257 (-2.04) | 0/10              | -1.30 $\pm$ 0.19 (0/10) |
| ViT-B    | +0.0012 | -0.0347 (-3.01) | 0/10              | -1.40 $\pm$ 0.25 (0/10) |
| ViT-L    | +0.0000 | -0.0337 (-3.73) | 0/10              | -1.61 $\pm$ 0.19 (0/10) |
| DINO-B   | +0.0002 | -0.0092 (-1.02) | 0/10              | +1.75 $\pm$ 0.16 (0/10) |
| DINOv2-S | +0.0007 | +0.0083 (+0.71) | 10/10             | +1.38 $\pm$ 0.14 (0/10) |
| DINOv2-B | +0.0018 | -0.0155 (-1.18) | 0/10              | +0.41 $\pm$ 0.11 (0/10) |
| DINOv2-L | -0.0007 | -0.0267 (-2.21) | 0/10              | -0.20 $\pm$ 0.14 (0/10) |
| DINOv2-G | -0.0021 | -0.0263 (-1.50) | 0/10              | -0.04 $\pm$ 0.07 (0/10) |
| CLIP-B   | +0.0012 | -0.0171 (-1.48) | 1/10              | -1.25 $\pm$ 0.09 (0/10) |
| CLIP-L   | +0.0018 | -0.0090 (-0.74) | 2/10              | -0.88 $\pm$ 0.22 (0/10) |
| SigLIP-B | +0.0024 | -0.0121 (-0.78) | 1/10              | -0.84 $\pm$ 0.18 (0/10) |

Table 9: (continued)

(b) Two ViT-B/16 supervised on ImageNet-1k leaf labels only, through the census protocol, against the IN-21k census backbone.

| model                          | IN excess ( $r$ ) | C100 excess ( $r$ ) | depth, Gaussian hubs ( $z$ ) | depth, Haar hubs ( $z$ ) |
|--------------------------------|-------------------|---------------------|------------------------------|--------------------------|
| DeiT-B, IN-1k                  | -0.0073 (200)     | -0.0257 (200)       | -0.0213 (-1.26)              | -0.0197 (-1.21)          |
| ViT-B, augreg, IN-1k           | -0.0070 (199)     | -0.0117 (187)       | -0.0249 (-3.03)              | -0.0290 (-2.65)          |
| ViT-B, augreg, IN-21k (census) | -0.0162 (200)     | +0.0057 (43)        | -0.0335 (-3.62)              | -0.0347 (-3.01)          |

Table 9: (continued)

(c) A hyperbolic backbone and its Euclidean twin under the census of record and the depth test.

|            |         | CIFAR-100 ( $K=20$ , $n=100$ : false alarms uncontrolled) |                                  |           | ImageNet ( $K=30$ , $n=1000$ : false alarms controlled) |                                  |           |
|------------|---------|-----------------------------------------------------------|----------------------------------|-----------|---------------------------------------------------------|----------------------------------|-----------|
| model      | input   | exc. ( $r$ )                                              | $\hat{\delta}_{99.9}$ nat./Eucl. | depth $z$ | exc. ( $r$ )                                            | $\hat{\delta}_{99.9}$ nat./Eucl. | depth $z$ |
| MERU ViT-S | images  | −0.020 (199)                                              | 0.117/0.117                      | −1.52     | −0.008° (186)                                           | 0.088/0.088                      | −1.54     |
| CLIP ViT-S | images  | −0.028 (200)                                              | 0.106/0.107                      | −3.95     | −0.006° (171)                                           | 0.110/0.088                      | −1.85     |
| MERU ViT-B | images  | −0.020 (200)                                              | 0.114/0.114                      | −2.16     | −0.010 (196)                                            | 0.087/0.087                      | −0.86     |
| CLIP ViT-B | images  | −0.030 (200)                                              | 0.105/0.103                      | −3.28     | −0.009° (187)                                           | 0.107/0.085                      | −0.99     |
| MERU ViT-L | images  | −0.012° (195)                                             | 0.118/0.118                      | −0.29     | −0.007° (184)                                           | 0.088/0.088                      | −0.52     |
| CLIP ViT-L | images  | −0.016 (199)                                              | 0.123/0.112                      | −1.68     | −0.010 (197)                                            | 0.104/0.084                      | −1.21     |
| MERU ViT-S | prompts | −0.022 (200)                                              | 0.086/0.086                      | −0.67     | −0.003° (155)                                           | 0.085/0.086                      | +0.02     |
| CLIP ViT-S | prompts | −0.023 (200)                                              | 0.085/0.082                      | −0.78     | +0.006° (15)                                            | 0.093/0.087                      | +0.82     |
| MERU ViT-B | prompts | −0.012 (200)                                              | 0.096/0.096                      | −0.70     | −0.001° (132)                                           | 0.089/0.090                      | +0.76     |
| CLIP ViT-B | prompts | −0.017 (200)                                              | 0.094/0.088                      | +0.31     | +0.008° (2)                                             | 0.100/0.092                      | +0.18     |
| MERU ViT-L | prompts | −0.006° (190)                                             | 0.102/0.102                      | +0.58     | +0.001° (75)                                            | 0.093/0.093                      | +0.52     |
| CLIP ViT-L | prompts | −0.012 (200)                                              | 0.098/0.092                      | +0.48     | +0.006° (5)                                             | 0.101/0.092                      | +0.39     |

Table 9: (continued) (c) MERU (Desai et al., 2023) embeds images and text on the Lorentz hyperboloid with an entailment objective; its released twin is a CLIP baseline with the same ViT, RedCaps data and recipe minus the hyperbolic lift and entailment loss. Both pass through the instrument unchanged: class centroids of the projected embeddings (MERU: space-like hyperboloid coordinates; CLIP: unit vectors), Euclidean distances; exc. ( $r$ ): excess over the Haar spectrum-matched null with the number of 200 replicates above the real value (°: not genuine under BH over the 24 cells); nat./Eucl.: the same statistic in the model’s own metric (Lorentz distance between tangent-space-mean centroids for MERU, angular for CLIP) against the Euclidean one; depth  $z$ : the anisotropic matched-star test of panel (a) (10 star seeds), validated at  $n=1000$  only. On ImageNet images the excess runs from −0.007 to −0.010 for MERU and from −0.006 to −0.010 for its twin across ViT-S/B/L; MERU’s Lorentz-metric  $\hat{\delta}_{99.9}$  differs from its Euclidean value by at most 0.0002; depth  $z$  runs from −1.54 to −0.52 for MERU and from −1.85 to −0.99 for CLIP, so neither is more hierarchical than a matched star and MERU is never ahead of its twin. Its embeddings sit in the near-flat regime of the hyperboloid: learned curvature  $c=0.10$ , spatial norm times  $\sqrt{c}$  at the median 0.258–0.279 (95th percentile 0.263–0.287), and Lorentz-to-Euclidean pairwise distance ratio 0.9972–0.9991 at the median (0.9988–0.9997 on the class centroids), so the control tests the training objective, not a curved geometry. MERU’s hierarchy is a generic→specific (text→image) partial order, not a class taxonomy. (d) Five epochs, AdamW, lr  $10^{-5}$  encoder and  $10^{-3}$  heads, batch 256, seed 0; centroids of the census subset; decoupling over 10 seeds (fraction certified in parentheses). The hierarchical model is certified on both frames and keeps firing once decoupled; the leaf-CE model and the frozen checkpoint are certified intact but do not fire once decoupled on the frame of record, and the frozen checkpoint fires in 7 of 10 decoupled seeds on the balanced frame. The pre-set criterion (leaf-CE and frozen not certified intact) is not met, because ViT-B is certified by alignment; the discriminating comparison is the decoupled one.

(d) A trained positive control: the census ViT-B/16 fine-tuned on the full ImageNet-1k training set with leaf cross-entropy alone and with an added hierarchical cross-entropy at the WordNet 30/6/2 cuts (weights 1/2/4), one seed, identical batches; census excess under the centered Haar null, depth test and decoupling control on both frames.

| ViT-B/16                          | excess ( $r$ ) | $z$ , WordNet-30 | dec. $z$ (cert.) | $z$ , balanced | dec. $z$ (cert.) |
|-----------------------------------|----------------|------------------|------------------|----------------|------------------|
| frozen (census)                   | −0.0162 (200)  | −3.62            | −1.40 (0.0)      | −2.82          | −2.10 (0.7)      |
| leaf cross-entropy                | −0.0243 (200)  | −2.45            | −0.72 (0.0)      | −2.09          | −1.38 (0.0)      |
| leaf + hierarchical cross-entropy | −0.0293 (200)  | −2.97            | −2.38 (1.0)      | −3.57          | −3.51 (1.0)      |

Table 9: (continued)

(e) Radial control: the depth test on the same clouds after L2-normalizing every centroid and after removing the radial component of every offset (its projection onto the hub direction), under both matched stars, with the decoupling control on each transformed cloud.

| model    | L2-normalized |              |                       | radial component removed |              |                       |  |
|----------|---------------|--------------|-----------------------|--------------------------|--------------|-----------------------|--|
|          | $z$ aniso     | $z$ Haar-hub | decoupled $z$ (cert.) | $z$ aniso                | $z$ Haar-hub | decoupled $z$ (cert.) |  |
| ViT-T    | +0.27         | -0.17        | -0.78 (0.0)           | -2.06                    | -2.07        | -1.74 (0.3)           |  |
| ViT-S    | -0.76         | -0.93        | -1.16 (0.0)           | -2.26                    | -2.08        | -1.25 (0.0)           |  |
| ViT-B    | -2.45         | -2.52        | -0.57 (0.0)           | -3.60                    | -3.57        | -1.40 (0.0)           |  |
| ViT-L    | -2.00         | -2.23        | -1.41 (0.1)           | -3.99                    | -3.47        | -1.47 (0.0)           |  |
| DINO-B   | +0.82         | +0.95        | +2.00 (0.0)           | -1.61                    | -1.89        | +1.91 (0.0)           |  |
| DINOv2-S | +1.27         | +1.38        | +1.28 (0.0)           | +0.61                    | +0.58        | +1.43 (0.0)           |  |
| DINOv2-B | -0.07         | -0.01        | +1.28 (0.0)           | -1.18                    | -1.19        | +0.46 (0.0)           |  |
| DINOv2-L | -0.46         | -0.50        | +0.89 (0.0)           | -2.59                    | -2.28        | -0.01 (0.0)           |  |
| DINOv2-G | -0.78         | -0.76        | +0.88 (0.0)           | -1.75                    | -1.66        | +0.11 (0.0)           |  |
| CLIP-B   | -1.42         | -0.93        | -1.38 (0.0)           | -1.53                    | -1.06        | -1.19 (0.0)           |  |
| CLIP-L   | -1.57         | -1.12        | -0.82 (0.0)           | -1.05                    | -0.89        | -0.92 (0.0)           |  |
| SigLIP-B | -1.13         | -0.88        | -0.51 (0.0)           | -1.12                    | -1.07        | -0.89 (0.0)           |  |

Table 9: (continued)

## B.8 HOW MUCH DEPTH CAN THE TEST SEE? POWER ON SYNTHETIC AND IMPLANTED HIERARCHIES



Figure 6: **A two-level implant on the real ImageNet clouds is missed at the real within-cluster spread and detected once the spread is shrunk.** Detection rate at  $z \leq -2$  against implant strength  $s$  with the real within-cluster spread (solid) and shrunk (dashed): at full strength 0.05 against 1.00, and 0.00 at  $s = 0$  with the real spread.

(a) Synthetic hierarchies at  $d=768$ : two- and three-level hierarchies (pooled) and pure stars, within/between noise ratios 0.1/0.3/0.6, isotropic and anisotropic clusters pooled, 5 seeds.

| frame                           | $n$  | $K$ | power @ ratio 0.1 | 0.3  | 0.6  | star $z \leq -2$ | star $z \geq +2$ |
|---------------------------------|------|-----|-------------------|------|------|------------------|------------------|
| top-level, isotropic star       | 100  | 6   | 0.00              | 0.00 | 0.00 | 0.00             | 0.17             |
| top-level, isotropic star       | 100  | 12  | 0.00              | 0.00 | 0.00 | 0.00             | 0.13             |
| top-level, isotropic star       | 100  | 20  | 0.00              | 0.00 | 0.00 | 0.00             | 0.17             |
| top-level, isotropic star       | 100  | 30  | 0.00              | 0.00 | 0.00 | 0.00             | 0.17             |
| top-level, isotropic star       | 1000 | 6   | 0.00              | 0.00 | 0.00 | 0.00             | 0.17             |
| top-level, isotropic star       | 1000 | 12  | 0.00              | 0.00 | 0.00 | 0.00             | 0.17             |
| top-level, isotropic star       | 1000 | 20  | 0.00              | 0.00 | 0.00 | 0.00             | 0.17             |
| top-level, isotropic star       | 1000 | 30  | 0.00              | 0.00 | 0.00 | 0.00             | 0.17             |
| leaf clusters, anisotropic star | 100  | 6   | 1.00              | 1.00 | 1.00 | 0.00             | 0.00             |
| leaf clusters, anisotropic star | 100  | 12  | 1.00              | 1.00 | 1.00 | 0.17             | 0.00             |
| leaf clusters, anisotropic star | 100  | 20  | 1.00              | 1.00 | 1.00 | 0.17             | 0.00             |
| leaf clusters, anisotropic star | 1000 | 6   | 1.00              | 1.00 | 1.00 | 0.00             | 0.00             |
| leaf clusters, anisotropic star | 1000 | 12  | 1.00              | 1.00 | 1.00 | 0.00             | 0.00             |
| leaf clusters, anisotropic star | 1000 | 20  | 1.00              | 1.00 | 0.90 | 0.00             | 0.00             |
| leaf clusters, anisotropic star | 1000 | 30  | 1.00              | 1.00 | 0.70 | 0.00             | 0.00             |

Table 10: **The depth test has full power on synthetic hierarchies at the leaf frame and none at the top-level frame used on real backbones.** (a) Power = fraction of hierarchy runs with  $z \leq -2$ ; star columns = false-alarm rates in each direction. Top-level frame: the test receives the  $K$  super-cluster labels, as the earlier isotropic test did on the real backbones; its hub null keeps every frame cluster intact and only rearranges the  $K$  hubs, so hierarchy below the frame is invisible by construction. Leaf frame: the test receives the finest cluster labels with the anisotropic matched star (configurations with more leaves than points are infeasible and omitted). Pre-set bar: power  $\geq 0.8$  for two- and three-level hierarchies at noise ratios  $\leq 0.3$  for both  $n$ , and false alarms  $\leq 5\%$  in each direction. Leaf frame: met at  $n=1000$  (power 1.00 at ratios  $\leq 0.3$  and 0.90 at 0.6; false alarms 0% and 0%), failed at  $n=100$ , where  $K \geq 12$  (under ten points per cluster) gives 17% false alarms at  $z \leq -2$ . Top-level frame: power 0.00, false alarms 16% pooled. (b) Six super-hubs of five, hub =  $s$ -super-hub +  $(1-s+0.4s)$ -own Gaussian draw, hub cloud rescaled to the real RMS radius; the depth test of Table 9 runs unchanged, 5 implant seeds per  $s$ . hits: implant seeds with  $z \leq -2$  (of 5);  $s^*$ : first  $s$  with  $\geq 4$  of 5; within/between: RMS of the within-cluster offsets over the point-weighted RMS of the hub displacements (1.3 to 3.9 on the real clouds, against 0.1–0.6 in the synthetic sweep); tight: the same implant after shrinking the offsets to within/between = 0.6 (2 seeds). Power (fraction of runs with  $z \leq -2$ ) by  $s$ :  $s=0$ : 0.00,  $s=0.25$ : 0.00,  $s=0.5$ : 0.00,  $s=0.75$ : 0.00,  $s=1$ : 0.05; false alarms at  $s=0$ : 0 of 60. Under the Haar-hub star the same implants give 0 of 60 false alarms at  $s=0$  and 1 of 60 detections at  $s=1$  (power 0.02). Balanced frame: the  $K=30$  agglomerative cut of the WordNet distance matrix (complete linkage) with the smallest variance of cluster sizes, chosen and logged before any depth run; sizes 1–154 classes against 1–316 for the frame of record. (c) Implanted alignment is detected in 36 of 60 runs at full strength (60%) and in 1 of 60 at zero; the pre-set bar of 0.8 power at full strength is not met, and the power is backbone-dependent: every seed for the supervised ViTs and CLIP-B, none for DINOv2-S/B/L.

(b) Implanted two-level trees on the real ImageNet centroids: the hub arrangement replaced at strength  $s$ , every within-cluster offset kept; hits = implant seeds with  $z \leq -2$  at  $s=0/0.5/1$ ,  $\bar{z}_1$  = mean  $z$  at  $s=1$ , w/b = within/between spread.

| model    | WordNet-30 frame of record, Gaussian-hub star |     |       |       |             |                          | Haar-hub star |             | balanced frame (pre-specified) |       |       |
|----------|-----------------------------------------------|-----|-------|-------|-------------|--------------------------|---------------|-------------|--------------------------------|-------|-------|
|          | real $z$                                      | w/b | hits  | $s^*$ | $\bar{z}_1$ | tight $z$ at $s=0/0.5/1$ | hits $s=0/1$  | $\bar{z}_1$ | real $z$                       | hits  | $s^*$ |
| ViT-T    | -1.48                                         | 1.5 | 0/0/0 | -     | +0.12       | -0.60/-1.85/-9.22        | 0/0           | +0.04       | -2.08                          | 0/0/1 | -     |
| ViT-S    | -2.36                                         | 1.9 | 0/0/0 | -     | -1.22       | -0.23/-3.45/-10.05       | 0/0           | -0.70       | -1.72                          | 0/0/0 | -     |
| ViT-B    | -3.62                                         | 2.0 | 0/0/0 | -     | +0.37       | -1.03/-3.92/-11.45       | 0/0           | +0.41       | -2.82                          | 0/0/0 | -     |
| ViT-L    | -4.04                                         | 1.9 | 0/0/0 | -     | -1.37       | -1.61/-4.37/-12.68       | 0/0           | -0.98       | -5.57                          | 0/0/0 | -     |
| DINO-B   | -0.94                                         | 2.1 | 0/0/0 | -     | -0.31       | -1.61/-5.14/-12.09       | 0/0           | -0.29       | -0.98                          | 0/0/0 | -     |
| DINOv2-S | +0.66                                         | 3.0 | 0/0/0 | -     | -0.79       | -0.19/-3.06/-11.48       | 0/0           | -0.79       | +0.12                          | 0/0/0 | -     |
| DINOv2-B | -1.19                                         | 3.7 | 0/0/0 | -     | +0.41       | -0.01/-3.07/-11.17       | 0/0           | +0.55       | -0.32                          | 0/0/0 | -     |
| DINOv2-L | -2.39                                         | 3.9 | 0/0/0 | -     | +0.20       | +0.21/-2.67/-11.40       | 0/0           | +0.21       | -1.70                          | 0/0/0 | -     |
| DINOv2-G | -1.78                                         | 3.9 | 0/0/0 | -     | +1.30       | +0.24/-3.89/-11.58       | 0/0           | +1.23       | -0.93                          | 0/0/0 | -     |
| CLIP-B   | -1.95                                         | 1.3 | 0/0/0 | -     | -1.21       | -1.67/-3.81/-11.38       | 0/0           | -0.72       | -1.29                          | 0/0/0 | -     |
| CLIP-L   | -0.83                                         | 1.5 | 0/0/3 | -     | -2.18       | -2.43/-5.76/-13.51       | 0/1           | -1.72       | -0.68                          | 0/0/0 | -     |
| SigLIP-B | -0.90                                         | 1.5 | 0/0/0 | -     | -1.55       | -2.09/-5.04/-11.03       | 0/0           | -1.07       | -1.12                          | 0/0/0 | -     |

Table 10: (continued)

(c) Implanted hub alignment on the real ImageNet clouds: from the decoupled cloud of the decoupling control, each cluster's principal axis rotated toward its hub direction by a fraction  $s$  of the angle (hubs and within-cluster spectra unchanged); detection rate at  $z \leq -2$  over 5 seeds per backbone and pooled.

| model                                  | $s=0$ | $s=0.25$ | $s=0.5$ | $s=0.75$ | $s=1$ |
|----------------------------------------|-------|----------|---------|----------|-------|
| ViT-T                                  | 0.2   | 0.8      | 0.8     | 1.0      | 1.0   |
| ViT-S                                  | 0.0   | 0.4      | 1.0     | 1.0      | 1.0   |
| ViT-B                                  | 0.0   | 0.8      | 1.0     | 1.0      | 1.0   |
| ViT-L                                  | 0.0   | 0.6      | 1.0     | 1.0      | 1.0   |
| DINO-B                                 | 0.0   | 0.0      | 0.0     | 0.0      | 0.8   |
| DINOv2-S                               | 0.0   | 0.0      | 0.0     | 0.0      | 0.0   |
| DINOv2-B                               | 0.0   | 0.0      | 0.0     | 0.0      | 0.0   |
| DINOv2-L                               | 0.0   | 0.0      | 0.0     | 0.2      | 0.0   |
| DINOv2-G                               | 0.0   | 0.0      | 0.0     | 0.2      | 0.4   |
| CLIP-B                                 | 0.0   | 0.0      | 1.0     | 1.0      | 1.0   |
| CLIP-L                                 | 0.0   | 0.2      | 0.2     | 0.4      | 0.4   |
| SigLIP-B                               | 0.0   | 0.0      | 1.0     | 0.8      | 0.6   |
| pooled (12 backbones $\times$ 5 seeds) | 0.02  | 0.23     | 0.50    | 0.55     | 0.60  |

Table 10: (continued)

(d) A deep hierarchy at the real noise level: synthetic clouds with ViT-L's real ImageNet spectrum and a three-level implanted hierarchy (nested 2/6/30 cuts of the frame) at ViT-L's real within/between ratio, a flat two-level control (30 iid hubs, same spectrum and ratio), 5 seeds each, and the WordNet Poincaré embeddings of Nickel and Kiela trained on the transitive closure of the tree over the 1000 ImageNet leaves, read with Euclidean distances on the ball coordinates.

| cloud                                 | seed | $d$  | excess ( $r$ ) | depth $z$ | decoupled $z$ (cert.)  |
|---------------------------------------|------|------|----------------|-----------|------------------------|
| three-level hierarchy, ViT-L spectrum | 0    | 1024 | -0.0081 (199)  | -1.80     | -4.82 $\pm$ 0.73 (1.0) |
| three-level hierarchy, ViT-L spectrum | 1    | 1024 | -0.0095 (199)  | -2.45     | -4.44 $\pm$ 0.58 (1.0) |
| three-level hierarchy, ViT-L spectrum | 2    | 1024 | -0.0090 (199)  | -2.38     | -4.19 $\pm$ 0.60 (1.0) |
| three-level hierarchy, ViT-L spectrum | 3    | 1024 | -0.0099 (199)  | -2.08     | -4.48 $\pm$ 0.67 (1.0) |
| three-level hierarchy, ViT-L spectrum | 4    | 1024 | -0.0090 (199)  | -2.20     | -4.28 $\pm$ 0.54 (1.0) |
| flat control, ViT-L spectrum          | 0    | 1024 | -0.0198 (200)  | -0.08     | -1.61 $\pm$ 0.27 (0.1) |
| flat control, ViT-L spectrum          | 1    | 1024 | -0.0202 (200)  | -0.32     | -1.61 $\pm$ 0.31 (0.1) |
| flat control, ViT-L spectrum          | 2    | 1024 | -0.0196 (200)  | -0.20     | -1.76 $\pm$ 0.37 (0.2) |
| flat control, ViT-L spectrum          | 3    | 1024 | -0.0197 (200)  | -0.69     | -1.74 $\pm$ 0.33 (0.2) |
| flat control, ViT-L spectrum          | 4    | 1024 | -0.0192 (200)  | -0.27     | -1.68 $\pm$ 0.39 (0.2) |
| WordNet Poincaré, $d=10$              | 0    | 10   | +0.0830 (0)    | +2.36     | +0.32 $\pm$ 0.58 (0.0) |
| WordNet Poincaré, $d=50$              | 0    | 50   | +0.0354 (0)    | +1.53     | -2.22 $\pm$ 0.65 (0.6) |

Table 10: (continued)

(e) Power per backbone for a deep hierarchy at the backbone's own noise level: the three-level synthetic hierarchy of (d) built with each backbone's real ImageNet spectrum and within/between ratio; depth test at  $K=30$  (10 star seeds) on the intact cloud and the decoupling control (10 seeds) on each.

| model    | within/between | seeds | power | mean $z$ | decoupled power | decoupled mean $z$ |
|----------|----------------|-------|-------|----------|-----------------|--------------------|
| ViT-T    | 1.55           | 5     | 0.00  | -0.48    | 0.60            | -2.17              |
| ViT-S    | 1.93           | 5     | 0.00  | -1.04    | 0.90            | -2.77              |
| ViT-B    | 1.96           | 5     | 0.00  | -1.74    | 1.00            | -4.21              |
| ViT-L    | 1.88           | 5     | 0.80  | -2.18    | 1.00            | -4.44              |
| DINO-B   | 2.12           | 5     | 0.00  | -1.21    | 0.28            | -1.64              |
| DINOv2-S | 3.00           | 20    | 0.00  | -0.98    | 0.01            | -0.90              |
| DINOv2-B | 3.71           | 20    | 0.45  | -1.93    | 0.83            | -2.44              |
| DINOv2-L | 3.88           | 20    | 0.95  | -2.66    | 1.00            | -3.19              |
| DINOv2-G | 3.92           | 20    | 1.00  | -2.88    | 1.00            | -3.54              |
| CLIP-B   | 1.33           | 5     | 0.20  | -1.69    | 1.00            | -5.00              |
| CLIP-L   | 1.49           | 5     | 1.00  | -2.83    | 1.00            | -4.98              |
| SigLIP-B | 1.54           | 5     | 1.00  | -2.49    | 1.00            | -4.86              |

Table 10: (continued)

(f) False alarms of the balanced frame: the flat control of (d), ViT-L's spectrum and ratio, 5 seeds, read on the balanced thirty-superclass frame intact and decoupled (10 seeds each), with the flat hubs assigned by the balanced frame itself or by the WordNet-30 frame of record.

| flat control                       | intact fired | mean $z$ | decoupled fired | mean $z$ | range          |
|------------------------------------|--------------|----------|-----------------|----------|----------------|
| matched hubs (balanced frame)      | 0/5          | -0.46    | 0/50            | -0.65    | +0.26 to -1.52 |
| mismatched hubs (WordNet-30 frame) | 5/5          | -3.13    | 38/50           | -2.33    | -1.56 to -3.75 |

Table 10: (continued)

#### B.9 IS THE ISLAND REAL? THE TREE MAP UNDER SIX CONFIGURATIONS

Threshold sensitivity of the selection criterion of §5.4: the selected configuration is stable for degeneracy bounds in  $[0.45, 0.50]$  on ImageNet and  $[0.40, 0.50]$  on CIFAR-100; tighter bounds select cosine-complete on ImageNet (gap 0.13) and cosine-Ward on CIFAR-100 (gap 0.56), and in every case the label-free gap is moderate and configuration-sensitive, never a categorical island. The criterion must be applied per dataset: the configuration it selects on DBpedia (Euclidean, average linkage, where nothing is degenerate) is exactly the degenerate naive configuration on ImageNet, so a criterion transferred across datasets would reintroduce the island.



Figure 7: **The island is an artifact of the naive cut.** Pairwise ARI between dendrogram cuts under (a) the naive configuration, Euclidean distances with average linkage, where the DINOv2 block agrees with the rest at 0.03, and (b) the selected one, cosine with average linkage, where it agrees at 0.38.

(a) Selection diagnostics and mean pairwise ARI at the reference cut.

| dataset   | configuration              | max frac. | CPCC  | mean pairwise ARI at the reference cut |                    |        |
|-----------|----------------------------|-----------|-------|----------------------------------------|--------------------|--------|
|           |                            |           |       | B/L/G vs block                         | S, DINO-B vs block | within |
| ImageNet  | euclid-average (deg.)      | 0.97      | 0.761 | 0.03                                   | 0.08               | 0.64   |
| ImageNet  | euclid-complete (deg.)     | 0.83      | 0.606 | 0.03                                   | 0.17               | 0.44   |
| ImageNet  | euclid-ward                | 0.48      | 0.464 | 0.15                                   | 0.38               | 0.53   |
| ImageNet  | cosine-average (selected)  | 0.40      | 0.691 | 0.38                                   | 0.35               | 0.48   |
| ImageNet  | cosine-complete            | 0.32      | 0.539 | 0.13                                   | 0.17               | 0.50   |
| ImageNet  | cosine-ward                | 0.37      | 0.464 | 0.17                                   | 0.39               | 0.55   |
| CIFAR-100 | euclid-average (deg.)      | 0.80      | 0.798 | 0.00                                   | 0.41               | 0.57   |
| CIFAR-100 | euclid-complete (deg.)     | 0.66      | 0.722 | 0.17                                   | 0.40               | 0.54   |
| CIFAR-100 | euclid-ward                | 0.33      | 0.534 | 0.37                                   | 0.65               | 0.63   |
| CIFAR-100 | cosine-average (deg.)      | 0.53      | 0.777 | 0.30                                   | 0.54               | 0.58   |
| CIFAR-100 | cosine-complete (selected) | 0.39      | 0.698 | 0.39                                   | 0.51               | 0.63   |
| CIFAR-100 | cosine-ward                | 0.14      | 0.584 | 0.56                                   | 0.62               | 0.61   |

Table 11: **The categorical island is a chaining artifact of the naive cut; a moderate gap remains under every admissible configuration and closes only in triplet agreement under cosine-average.** (a) Six configurations (metric  $\times$  linkage) per dataset with the two quantities that drive the selection criterion of §5.4: the degeneracy diagnostic (largest-cluster fraction at the reference cut, worst model) and the mean cophenetic fidelity (CPCC) of each model’s dendrogram to its own distance matrix; configurations with max-cluster fraction  $> 0.5$  are excluded and, among the rest, the highest CPCC selects the marked row; then the mean pairwise ARI between the cuts at 30 (ImageNet) or 20 (CIFAR-100) clusters. (b) Dendrograms rebuilt from the census cache: ARI at the cut, Pearson correlation between the two trees’ cophenetic distance vectors, and agreement on which pair merges first over  $10^4$  random class triplets. Under the naive ImageNet configuration, cross-family ARI between the supervised ViTs and the contrastive VLMs reaches 0.82 while every DINOv2-vs-block pair stays at or below 0.06, and the worst model’s largest cluster holds 97% of the classes. Under the selected configuration the DINOv2 family agrees with the block at 0.38 against 0.48 within the block. Under the other 3 admissible ImageNet configurations the DINOv2-vs-block agreement is 0.13–0.17 against 0.50–0.55 within the block; over the admissible configurations and the three measures the median fraction of within-block agreement missing is 0.34, about a third. (c) Two resamples of the same model agree on 0.90 of the triplets on average, 0.85 for the lowest backbone: the ceiling against which the cross-model 0.77 and the within-block 0.76 are read.

(b) Three measures of DINOv2-B/L/G against the supervised+contrastive block, each written as vs block / within block.

| dataset   | configuration              | ARI at the cut | cophenetic corr. | triplet agreement |
|-----------|----------------------------|----------------|------------------|-------------------|
| ImageNet  | euclid-average (deg.)      | 0.03 / 0.64    | 0.36 / 0.80      | 0.47 / 0.74       |
| ImageNet  | euclid-complete (deg.)     | 0.03 / 0.44    | 0.20 / 0.66      | 0.41 / 0.65       |
| ImageNet  | euclid-ward                | 0.16 / 0.53    | 0.62 / 0.84      | 0.67 / 0.85       |
| ImageNet  | cosine-average (selected)  | 0.39 / 0.48    | 0.48 / 0.78      | 0.77 / 0.76       |
| ImageNet  | cosine-complete            | 0.13 / 0.49    | 0.18 / 0.72      | 0.36 / 0.71       |
| ImageNet  | cosine-ward                | 0.17 / 0.55    | 0.59 / 0.85      | 0.69 / 0.86       |
| CIFAR-100 | euclid-average (deg.)      | 0.00 / 0.57    | 0.15 / 0.78      | 0.41 / 0.74       |
| CIFAR-100 | euclid-complete (deg.)     | 0.17 / 0.54    | 0.23 / 0.72      | 0.41 / 0.66       |
| CIFAR-100 | euclid-ward                | 0.37 / 0.63    | 0.63 / 0.77      | 0.70 / 0.79       |
| CIFAR-100 | cosine-average (deg.)      | 0.30 / 0.58    | 0.42 / 0.72      | 0.51 / 0.64       |
| CIFAR-100 | cosine-complete (selected) | 0.39 / 0.63    | 0.46 / 0.64      | 0.50 / 0.61       |
| CIFAR-100 | cosine-ward                | 0.56 / 0.61    | 0.74 / 0.73      | 0.76 / 0.76       |

Table 11: (continued)

(c) Within-model ceiling under the selected ImageNet configuration: the three measures of (b) between pairs of the 30 bootstrap centroid sets of each backbone (mean  $\pm$  s.d. over the 435 pairs) and the lowest triplet agreement over pairs.

| model               | ARI at the cut  | cophenetic corr. | triplet agreement | lowest triplet |
|---------------------|-----------------|------------------|-------------------|----------------|
| ViT-T               | $0.81 \pm 0.03$ | $0.94 \pm 0.02$  | $0.93 \pm 0.04$   | 0.85           |
| ViT-S               | $0.80 \pm 0.11$ | $0.94 \pm 0.01$  | $0.88 \pm 0.08$   | 0.70           |
| ViT-B               | $0.94 \pm 0.01$ | $0.94 \pm 0.01$  | $0.86 \pm 0.04$   | 0.75           |
| ViT-L               | $0.84 \pm 0.06$ | $0.96 \pm 0.01$  | $0.96 \pm 0.02$   | 0.87           |
| DINO-B              | $0.78 \pm 0.10$ | $0.94 \pm 0.01$  | $0.96 \pm 0.01$   | 0.95           |
| DINOv2-S            | $0.66 \pm 0.05$ | $0.91 \pm 0.01$  | $0.86 \pm 0.05$   | 0.69           |
| DINOv2-B            | $0.64 \pm 0.09$ | $0.93 \pm 0.01$  | $0.86 \pm 0.03$   | 0.77           |
| DINOv2-L            | $0.78 \pm 0.05$ | $0.94 \pm 0.01$  | $0.85 \pm 0.04$   | 0.71           |
| DINOv2-G            | $0.76 \pm 0.05$ | $0.95 \pm 0.01$  | $0.88 \pm 0.02$   | 0.84           |
| CLIP-B              | $0.87 \pm 0.04$ | $0.94 \pm 0.01$  | $0.93 \pm 0.02$   | 0.88           |
| CLIP-L              | $0.82 \pm 0.06$ | $0.97 \pm 0.01$  | $0.95 \pm 0.01$   | 0.92           |
| SigLIP-B            | $0.79 \pm 0.07$ | $0.95 \pm 0.02$  | $0.91 \pm 0.03$   | 0.84           |
| mean over backbones | $0.79 \pm 0.08$ | $0.94 \pm 0.01$  | $0.90 \pm 0.04$   | 0.85           |

Table 11: (continued)

## B.10 WHOSE TAXONOMY? WORDNET ALIGNMENT, SUPERCLASS RECOVERY, DBPEDIA AND HIERARCAPS

On HierarCaps (Alper & Averbuch-Elor, 2024) (1000 four-level caption chains), text encoders show the same reading with no class set: the projection radius orders levels within a chain, sibling triplets are resolved at 0.83–0.97 (chance 0.5), and the leaf excess repeats the text census (Table 12).

(a) Alignment with WordNet on ImageNet.

| model                 | Spearman $\rho$ , inter-centroid vs WordNet distance |         |          |              | raw $\hat{\delta}$ of WordNet / random groups |
|-----------------------|------------------------------------------------------|---------|----------|--------------|-----------------------------------------------|
|                       | $\rho_{WN}$                                          | shuffle | Gaussian | single image |                                               |
| ViT-T                 | +0.52                                                | +0.02   | +0.01    | –            | 0.146 / 0.159                                 |
| ViT-S                 | +0.50                                                | +0.00   | –0.00    | –            | 0.110 / 0.130                                 |
| ViT-B                 | +0.49                                                | +0.03   | +0.01    | –            | 0.092 / 0.120                                 |
| ViT-L                 | +0.53                                                | +0.01   | +0.03    | +0.25        | 0.104 / 0.117                                 |
| DINO-B                | +0.36                                                | –0.01   | –0.01    | –            | 0.084 / 0.117                                 |
| DINOv2-S              | +0.22                                                | +0.01   | +0.01    | –            | 0.077 / 0.100                                 |
| DINOv2-B              | +0.18                                                | –0.01   | +0.01    | –            | 0.068 / 0.069                                 |
| DINOv2-L              | +0.18                                                | –0.00   | –0.03    | +0.08        | 0.060 / 0.058                                 |
| DINOv2-G              | +0.20                                                | –0.03   | –0.01    | +0.03        | 0.054 / 0.056                                 |
| CLIP-B                | +0.57                                                | +0.00   | +0.01    | –            | 0.126 / 0.174                                 |
| CLIP-L                | +0.59                                                | +0.01   | +0.03    | +0.26        | 0.117 / 0.162                                 |
| SigLIP-B              | +0.58                                                | –0.01   | +0.01    | –            | 0.118 / 0.158                                 |
| DeiT-B (IN-1k)        | +0.08                                                | +0.01   | –        | –            | –                                             |
| ViT-B (augreg, IN-1k) | +0.52                                                | +0.01   | –        | –            | –                                             |

Table 12: **Alignment with the human taxonomy follows supervision and recipe, the superclasses are recovered by every family, and the recovery is not WordNet circularity.** (a)  $\rho_{WN}$ : Spearman correlation between inter-centroid and WordNet tree distances on ImageNet, with the label-shuffle and Gaussian-cloud controls; single image: the same correlation on single-image distances (one image per class); WordNet / random groups: raw  $\hat{\delta}$  of WordNet-coherent against random class groupings. (b) ARI of the 20-cluster cut against the true CIFAR-100 superclasses per configuration (degenerate configurations marked; the criterion-selected one is cosine-complete). Paired within each of the 3 admissible configurations, the label-supervised ViT-B recovers more than each DINOv2 model in 11 of 12 comparisons; DINOv2 itself recovers the superclasses at ARI 0.19–0.52, and the best cut over models and configurations reaches 0.61 against zero for permuted labels. Circularity controls: WordNet-coherent groupings read a lower raw  $\delta$  than random ones in 11 of 12 models, and single-image distances still correlate with WordNet at up to +0.26. The two leaf-label ViT-B/16 of Table 9: the augreg recipe is aligned with WordNet like the IN-21k ViTs ( $\rho_{WN}$  +0.52) while DeiT-B is nearly unaligned (+0.08) yet recovers the CIFAR-100 superclasses at ARI up to 0.56, so WordNet alignment is recipe-dependent even among leaf-supervised ViTs.

(b) Recovery of the CIFAR-100 superclasses: ARI of the 20-cluster cut per configuration ( $\dagger$ : degenerate on CIFAR-100, euclid-average being the naive configuration;  $*$ : criterion-selected).

| model                 | euclid-avg $\dagger$ | euclid-ward | cosine-avg $\dagger$ | cosine-complete $*$ | cosine-ward |
|-----------------------|----------------------|-------------|----------------------|---------------------|-------------|
| ViT-T                 | 0.31                 | 0.38        | 0.29                 | 0.33                | 0.33        |
| ViT-S                 | 0.35                 | 0.56        | 0.37                 | 0.47                | 0.56        |
| ViT-B                 | 0.39                 | 0.61        | 0.43                 | 0.45                | 0.57        |
| ViT-L                 | 0.38                 | 0.55        | 0.49                 | 0.48                | 0.60        |
| DINO-B                | 0.22                 | 0.48        | 0.28                 | 0.38                | 0.54        |
| DINOv2-S              | 0.12                 | 0.51        | 0.32                 | 0.33                | 0.49        |
| DINOv2-B              | 0.02                 | 0.47        | 0.35                 | 0.42                | 0.51        |
| DINOv2-L              | 0.01                 | 0.32        | 0.25                 | 0.50                | 0.48        |
| DINOv2-G              | 0.00                 | 0.19        | 0.08                 | 0.20                | 0.52        |
| CLIP-B                | 0.18                 | 0.48        | 0.18                 | 0.37                | 0.48        |
| CLIP-L                | 0.35                 | 0.52        | 0.32                 | 0.41                | 0.54        |
| SigLIP-B              | 0.22                 | 0.49        | 0.22                 | 0.37                | 0.52        |
| DeiT-B (IN-1k)        | 0.22                 | 0.56        | 0.23                 | 0.38                | 0.54        |
| ViT-B (augreg, IN-1k) | 0.30                 | 0.44        | 0.40                 | 0.42                | 0.48        |

Table 12: (continued)

(c) DBpedia Classes (219 leaf classes, 3 levels), a text hierarchy independent of WordNet; gains in pp.

| model    | supremum, Haar        |        | census of record      |        |             | trip. NC H-R FS H-R |       |                  |
|----------|-----------------------|--------|-----------------------|--------|-------------|---------------------|-------|------------------|
|          | $\hat{\delta}_{\max}$ | excess | $\hat{\delta}_{99.9}$ | excess | $r(p)$      |                     |       |                  |
| BGE-base | 0.122                 | -0.032 | 0.082                 | -0.021 | 200 (0.005) | 0.89                | -0.59 | +0.05 $\pm$ 0.05 |
| E5-base  | 0.127                 | -0.028 | 0.086                 | -0.017 | 200 (0.005) | 0.89                | -0.43 | +0.04 $\pm$ 0.03 |
| GTE-base | 0.132                 | -0.024 | 0.088                 | -0.017 | 200 (0.005) | 0.89                | -0.82 | +0.08 $\pm$ 0.06 |

Table 12: (continued) (c) Excess under the original supremum reading (3 Gaussian replicates) and under the census of record (Haar null, 99.9th-percentile statistic, 200 replicates;  $r$  = replicates above the real value, left-tail  $p$ ), sibling-triplet agreement, and the metric gains, marginal as the diagnostic predicts. Tree map on the same classes: no configuration degenerates (largest cluster  $\leq 10\%$ ), the embedders recover the true 70-way level-2 partition at ARI 0.21–0.29 in all 6 configurations and agree with each other at cross-model ARI 0.65–0.82; the configuration the criterion selects there (Euclidean, average linkage) is the one that degenerates on ImageNet, which is why the criterion is applied per dataset. (d) Radial ordering of the four levels under the paper’s projection (chance for strict monotonicity =  $1/24 \approx 4.2\%$ ; shuffle controls  $\approx 0$ ), sibling triplet agreement (cosine; chance 0.5), and leaf  $\hat{\delta}$  with spectrum-null excess.

(d) HierarCaps (1000 four-level caption chains): a multi-level text hierarchy with no class set.

| model    | $\rho(\text{level, radius})$ | % monotone | trip. L1 | trip. L2 | $\hat{\delta}$ leaves | excess |
|----------|------------------------------|------------|----------|----------|-----------------------|--------|
| BGE-base | +0.80 $\pm$ 0.31             | 49.6       | 0.88     | 0.97     | 0.130                 | +0.003 |
| E5-base  | +0.87 $\pm$ 0.19             | 53.9       | 0.90     | 0.97     | 0.139                 | +0.010 |
| GTE-base | +0.47 $\pm$ 0.55             | 24.0       | 0.90     | 0.97     | 0.135                 | +0.006 |
| CLIP-B   | +0.64 $\pm$ 0.43             | 27.2       | 0.86     | 0.95     | 0.123                 | -0.011 |
| CLIP-L   | +0.51 $\pm$ 0.49             | 18.7       | 0.86     | 0.96     | 0.117                 | -0.010 |
| SigLIP-B | +0.32 $\pm$ 0.55             | 9.8        | 0.83     | 0.91     | 0.138                 | +0.014 |

Table 12: (continued)

## B.11 IS LOCAL STRUCTURE SHARED? PERMUTATION-CALIBRATED AGREEMENT

| measure                          | raw   | null mean | $\tau_{0.05}$ | calibrated | null-centered | pairs $p < 0.05$ |
|----------------------------------|-------|-----------|---------------|------------|---------------|------------------|
| mutual-kNN ( $k=10$ ), Euclidean | 0.472 | 0.010     | 0.015         | 0.464      | 0.462         | 66/66            |
| mutual-kNN, Poincaré             | 0.425 | 0.010     | 0.019         | 0.414      | 0.415         | 66/66            |
| linear CKA, Euclidean            | 0.633 | 0.108     | 0.111         | 0.577      | 0.524         | 66/66            |
| linear CKA, Poincaré             | 0.640 | 0.108     | 0.111         | 0.586      | 0.532         | 66/66            |

Table 13: **Cross-model agreement survives permutation calibration.** Means over the 66 model pairs of the 12 vision backbones on the 1000 ImageNet class centroids (census cache). Calibration of Gröger et al. (2026) with  $K=200$  permutations of the class correspondence and  $\alpha=0.05$ , scalar (no layer search):  $\tau_{0.05}$  is the  $\lceil 0.95(K+1) \rceil$ -th order statistic of the observed score and its nulls (eq. 9),  $p = (1 + \#\{\text{null} \geq \text{obs}\})/(K+1)$  (eq. 10), calibrated =  $\max\{(\text{obs} - \tau_{0.05})/(1 - \tau_{0.05}), 0\}$  (eq. 12); “null-centered” is the earlier raw-minus-null-mean variant, kept for continuity. The mKNN null mean equals the analytic chance level  $k/(n-1) = 10/999$ .

#### B.12 WHAT FOLLOWS FOR PRACTICE? THE COROLLARY

**One parameter-free projection and two tasks serve the corollary.** The parameter-free map  $\phi(x) = \tanh(\frac{1}{2}\|s(x - \mu)\|) \cdot \frac{s(x - \mu)}{\|s(x - \mu)\|}$  centers features, scales the 95th-percentile norm to radius  $1/\sqrt{2}$ , and preserves directions; prototypes in the ball are means of projected points re-mapped through the origin. Tasks: nearest-centroid (NC) and 5-way 5-shot (FS, 1000 paired episodes); metric arms: Euclidean (R), Poincaré (H), the radial map with Euclidean distances (RT), cosine (COS), and the L2-normalized stack. Changing the readout metric on the intact configuration, which is what the rule selects among, is distinct from deforming the configuration itself.

**Both readings predict the gain within datasets.** Within each hierarchical dataset the Poincaré advantage tracks the record excess: the nearest-centroid correlation is  $-0.73$  on ImageNet with a confidence interval that excludes zero on all four datasets, the few-shot correlation excludes zero on CIFAR-100, CIFAR-10 and DTD but not on ImageNet, and the depth-test  $z$  predicts no gain anywhere (Table 14). The raw reading correlates as strongly ( $-0.84$  on ImageNet for nearest-centroid) and survives partialing out dimension, baseline accuracy, family and spectral covariates; without DINOv2 it holds on ImageNet and CIFAR-100 but not on CIFAR-10/DTD. Pooled cross-dataset correlations are family-driven and not used as evidence; on flat-label datasets no metric helps. Gains carry paired confidence intervals and a McNemar test.

**The training objective picks the metric.** The radial map with Euclidean distances changes nothing, so the gain is the metric itself, and the split follows the objective: cosine suffices for DINOv2, Poincaré  $\approx$  cosine for supervised ViTs, and Poincaré beats cosine for CLIP by up to  $+1.3\text{pp}$  even after L2 normalization (Table 14). The rule is read off the families it is scored on: it describes them rather than validating a policy.

**A diagnostic, not a policy.** Khrulkov et al. (2020) estimate  $c \approx 0.33$  from the raw reading of their datasets; the same rule on structureless Gaussian clouds gives the curvatures of Section 6. Cosine is a safe default, gating on the reading is the weakest policy tested, and gains are modest and limited to prototype-based tasks (sample-level tasks never benefit, Table 8). On DBpedia, a three-level text hierarchy, the diagnostic predicts correctly on both sides: genuine excess, no metric gain.

(a) Task accuracies and zero-cost metric gains per cell (10 backbones with cached per-dataset features  $\times$  6 datasets).

| model    | data   | NC acc. (%) |        | FS acc. (%) |        | FS advantage over Euclidean (pp) |            |                           | McNemar        |
|----------|--------|-------------|--------|-------------|--------|----------------------------------|------------|---------------------------|----------------|
|          |        | Eucl.       | Poinc. | Eucl.       | Poinc. | cosine                           | radial map | Poinc. <sub>N</sub> - cos | NC H vs R      |
| ViT-T    | IN     | 68.1        | 67.2   | 97.5        | 97.6   | -0.01                            | +0.01      | +0.08                     | $10^{-22}$ (-) |
|          | C100   | 29.3        | 28.2   | 62.4        | 63.1   | +1.01                            | +0.45      | +0.55                     | $10^{-5}$ (-)  |
|          | C10    | 87.5        | 87.4   | 87.6        | 88.1   | +0.66                            | +0.19      | +0.29                     | 0.19 (-)       |
|          | DTD    | 63.7        | 62.5   | 83.3        | 83.9   | -0.23                            | +0.32      | +0.69                     | 0.03 (-)       |
|          | FMNIST | 78.5        | 78.4   | 81.6        | 81.9   | +0.21                            | +0.10      | +0.31                     | 0.38 (-)       |
|          | MNIST  | 75.9        | 74.0   | 74.3        | 74.6   | +0.64                            | +0.52      | +0.20                     | $10^{-13}$ (-) |
| ViT-S    | IN     | 77.7        | 77.5   | 98.6        | 98.7   | +0.17                            | -0.02      | -0.07                     | 0.09 (-)       |
|          | C100   | 74.9        | 74.6   | 92.2        | 92.4   | +0.59                            | +0.16      | -0.00                     | 0.10 (-)       |
|          | C10    | 93.4        | 93.2   | 92.0        | 92.5   | +0.56                            | +0.27      | +0.19                     | 0.02 (-)       |
|          | DTD    | 69.3        | 68.5   | 85.4        | 86.0   | -0.03                            | +0.25      | +0.73                     | 0.10 (-)       |
|          | FMNIST | 80.7        | 80.4   | 83.7        | 83.5   | +0.28                            | +0.08      | -0.19                     | 0.08 (-)       |
|          | MNIST  | 84.8        | 84.1   | 80.5        | 81.4   | +0.34                            | +0.65      | +0.84                     | $10^{-4}$ (-)  |
| ViT-B    | IN     | 81.6        | 81.7   | 98.9        | 99.0   | +0.20                            | -0.04      | -0.05                     | 0.21 (+)       |
|          | C100   | 77.3        | 77.0   | 92.5        | 92.7   | +0.56                            | +0.12      | -0.05                     | 0.07 (-)       |
|          | C10    | 94.5        | 94.0   | 92.7        | 93.0   | +0.49                            | +0.18      | +0.15                     | $10^{-5}$ (-)  |
|          | DTD    | 70.8        | 70.8   | 85.4        | 86.2   | +0.28                            | +0.19      | +0.69                     | 1.00 (-)       |
|          | FMNIST | 81.3        | 81.6   | 83.4        | 83.3   | +0.03                            | +0.09      | -0.04                     | 0.08 (+)       |
|          | MNIST  | 81.6        | 80.5   | 76.8        | 77.6   | +0.19                            | +0.51      | +0.66                     | $10^{-7}$ (-)  |
| ViT-L    | IN     | 86.1        | 86.2   | 99.4        | 99.4   | +0.11                            | -0.03      | -0.02                     | 0.58 (+)       |
|          | C100   | 85.3        | 85.2   | 94.8        | 95.5   | +1.17                            | +0.31      | +0.12                     | 0.59 (-)       |
|          | C10    | 97.5        | 97.5   | 95.0        | 95.8   | +1.25                            | +0.35      | +0.59                     | 0.76 (+)       |
|          | DTD    | 71.8        | 71.5   | 85.6        | 86.5   | +0.13                            | +0.16      | +1.02                     | 0.69 (-)       |
|          | FMNIST | 83.7        | 83.5   | 84.4        | 84.7   | +0.00                            | +0.17      | +0.23                     | 0.12 (-)       |
|          | MNIST  | 84.2        | 83.6   | 80.8        | 81.6   | +0.08                            | +0.42      | +0.67                     | 0.00 (-)       |
| DINOv2-S | IN     | 74.7        | 74.4   | 97.4        | 97.9   | +0.68                            | +0.03      | -0.22                     | 0.00 (-)       |
|          | C100   | 73.4        | 74.4   | 94.3        | 95.1   | +0.80                            | +0.15      | +0.06                     | $10^{-6}$ (+)  |
|          | C10    | 95.0        | 95.2   | 92.9        | 93.5   | -0.36                            | +0.13      | +1.06                     | 0.14 (+)       |
|          | DTD    | 73.4        | 73.4   | 89.8        | 90.5   | +0.30                            | +0.04      | +0.36                     | 1.00 (+)       |
|          | FMNIST | 82.2        | 82.2   | 85.4        | 85.5   | +0.02                            | +0.07      | +0.07                     | 1.00 (-)       |
|          | MNIST  | 83.4        | 81.9   | 80.2        | 80.8   | -0.07                            | +0.45      | +0.67                     | $10^{-11}$ (-) |
| DINOv2-B | IN     | 78.9        | 78.9   | 97.0        | 97.8   | +1.40                            | +0.04      | -0.63                     | 0.57 (+)       |
|          | C100   | 87.4        | 87.9   | 96.5        | 97.2   | +1.11                            | +0.09      | -0.34                     | $10^{-4}$ (+)  |
|          | C10    | 97.1        | 97.4   | 94.7        | 95.4   | +0.27                            | +0.03      | +0.43                     | $10^{-4}$ (+)  |
|          | DTD    | 74.2        | 74.7   | 89.2        | 90.3   | +1.48                            | +0.01      | -0.35                     | 0.31 (+)       |
|          | FMNIST | 83.2        | 82.8   | 85.9        | 85.9   | +0.21                            | +0.04      | -0.18                     | 0.01 (-)       |
|          | MNIST  | 80.6        | 78.8   | 76.6        | 77.1   | +0.17                            | +0.48      | +0.47                     | $10^{-16}$ (-) |
| DINOv2-L | IN     | 79.8        | 80.4   | 96.2        | 97.1   | +2.13                            | +0.02      | -1.24                     | $10^{-14}$ (+) |
|          | C100   | 85.6        | 87.7   | 96.1        | 97.1   | +1.80                            | +0.09      | -0.74                     | $10^{-38}$ (+) |
|          | C10    | 97.9        | 98.3   | 94.1        | 95.1   | +0.92                            | -0.00      | +0.11                     | $10^{-8}$ (+)  |
|          | DTD    | 74.4        | 75.2   | 87.1        | 88.4   | +2.40                            | +0.03      | -0.98                     | 0.11 (+)       |
|          | FMNIST | 84.4        | 84.5   | 87.8        | 87.9   | +0.29                            | +0.03      | -0.03                     | 0.40 (+)       |
|          | MNIST  | 74.1        | 72.7   | 70.5        | 71.1   | +0.25                            | +0.43      | +0.46                     | $10^{-12}$ (-) |
| DINOv2-G | IN     | 78.4        | 78.3   | 96.2        | 97.1   | +2.57                            | +0.03      | -1.65                     | 0.10 (-)       |
|          | C100   | 89.8        | 91.0   | 95.9        | 97.2   | +2.44                            | +0.13      | -1.10                     | $10^{-16}$ (+) |
|          | C10    | 97.8        | 98.2   | 92.3        | 93.8   | +2.51                            | -0.02      | -1.01                     | $10^{-13}$ (+) |
|          | DTD    | 72.9        | 73.1   | 87.2        | 88.5   | +2.63                            | +0.01      | -1.38                     | 0.81 (+)       |
|          | FMNIST | 85.9        | 85.8   | 89.0        | 89.2   | +0.38                            | +0.04      | -0.07                     | 0.72 (-)       |
|          | MNIST  | 79.3        | 78.6   | 72.9        | 73.8   | +0.30                            | +0.56      | +0.72                     | 0.00 (-)       |
| CLIP-B   | IN     | 61.4        | 59.9   | 96.8        | 96.8   | -0.04                            | -0.00      | +0.14                     | $10^{-40}$ (-) |
|          | C100   | 65.6        | 65.1   | 87.7        | 88.9   | -0.03                            | +0.44      | +1.21                     | 0.07 (-)       |
|          | C10    | 87.9        | 88.3   | 89.5        | 90.1   | -0.19                            | +0.20      | +0.90                     | $10^{-4}$ (+)  |
|          | DTD    | 63.7        | 63.2   | 82.2        | 83.1   | -0.23                            | +0.36      | +1.29                     | 0.46 (-)       |
|          | FMNIST | 75.9        | 75.1   | 78.8        | 78.9   | +0.06                            | +0.17      | +0.10                     | $10^{-4}$ (-)  |
|          | MNIST  | 70.1        | 66.6   | 70.9        | 69.7   | +0.04                            | +0.23      | -1.17                     | $10^{-45}$ (-) |
| CLIP-L   | IN     | 75.0        | 74.5   | 98.4        | 98.6   | +0.01                            | +0.00      | +0.22                     | $10^{-8}$ (-)  |
|          | C100   | 74.6        | 74.5   | 90.0        | 90.8   | +0.12                            | +0.39      | +1.23                     | 0.66 (-)       |
|          | C10    | 95.9        | 95.9   | 92.4        | 93.0   | +0.16                            | +0.25      | +0.91                     | 0.59 (-)       |
|          | DTD    | 68.6        | 69.0   | 85.2        | 86.2   | -0.08                            | +0.36      | +1.24                     | 0.45 (+)       |
|          | FMNIST | 76.2        | 75.9   | 76.8        | 77.3   | +0.12                            | +0.33      | +0.68                     | 0.09 (-)       |
|          | MNIST  | 85.2        | 86.1   | 83.7        | 84.7   | +0.08                            | +0.58      | +1.06                     | $10^{-7}$ (+)  |

Table 14: **Both readings predict the zero-cost gain within datasets, the depth verdict predicts nothing, and cosine is the safe default.** (a) Nearest-centroid (NC) and 5-way 5-shot (FS, 1000 paired episodes) accuracy under Euclidean and Poincaré readouts, regenerated from the audited re-runs; FS advantage of cosine and of the radial map with Euclidean distances (the map without the metric changes nothing) over Euclidean, and of the Poincaré readout over cosine after L2 normalization; McNemar  $p$  for the NC test-set decisions, Poincaré vs Euclidean (order of magnitude when  $p < 10^{-3}$ ; sign of the NC advantage in parentheses).

(b) Correlation between the raw supremum  $\delta_{\text{norm}}$  and the metric gain within each dataset ( $n=10$  backbones), with bootstrap CI95 and the family-identity control (demeaned: both variables centered within family; within-family  $r$  for ViT and DINOv2, sign only for CLIP).

| gain      | dataset                    | $r$ [CI95]           | demeaned | ViT   | DINOv2 | CLIP |
|-----------|----------------------------|----------------------|----------|-------|--------|------|
| NC H—R    | ImageNet                   | −0.83 [−0.97, −0.65] | −0.73    | −0.99 | −0.77  | (−1) |
|           | CIFAR-100                  | −0.87 [−0.97, −0.78] | −0.45    | −0.68 | −0.33  | (−1) |
|           | CIFAR-10                   | −0.63 [−0.97, −0.06] | −0.31    | +0.01 | −0.98  | (+1) |
|           | DTD                        | −0.46 [−0.90, +0.22] | −0.57    | −0.78 | −0.76  | (+1) |
|           | pooled (cluster bootstrap) | −0.45 [−0.60, −0.36] | —        | —     | —      | —    |
| FS H—R    | ImageNet                   | −0.57 [−0.91, +0.14] | −0.73    | −0.04 | −0.97  | (−1) |
|           | CIFAR-100                  | −0.67 [−0.95, −0.15] | −0.63    | −0.41 | −0.96  | (+1) |
|           | CIFAR-10                   | −0.78 [−0.93, −0.23] | −0.64    | −0.22 | −0.79  | (−1) |
|           | DTD                        | −0.71 [−0.94, −0.32] | −0.86    | −0.95 | −0.98  | (+1) |
|           | pooled (cluster bootstrap) | −0.18 [−0.50, +0.12] | —        | —     | —      | —    |
| FS best—R | ImageNet                   | −0.64 —              | −0.71    | −0.57 | −0.90  | (−1) |
|           | CIFAR-100                  | −0.83 —              | −0.82    | −0.58 | −0.96  | (+1) |
|           | CIFAR-10                   | −0.63 —              | −0.55    | −0.26 | −0.65  | (−1) |
|           | DTD                        | −0.60 —              | −0.70    | −0.95 | −0.88  | (+1) |

Table 14: (continued) (b) Within-dataset correlations of the raw supremum reading; bootstrap CI95 from 10k resamples of the 10 models, pooled rows by cluster bootstrap over datasets; family control: both variables centered within family before correlating, and within-family correlations (ViT, DINOv2: 4 models each; CLIP has two, so only its sign is shown). Without DINOv2 (six backbones) the raw NC correlation holds on ImageNet and CIFAR-100 (−0.87/−0.65) but not on CIFAR-10/DTD (+0.23/−0.34); pooled cross-dataset correlations of the best-metric advantage do not survive the family control (−0.31 → −0.18 over all 60 cells, −0.33 → −0.04 on hierarchical cells) and are family-driven. On the record excess the nearest-centroid prediction survives (CI95 excluding zero) on ImageNet, CIFAR-100, CIFAR-10, DTD and the few-shot prediction on CIFAR-100, CIFAR-10, DTD; the depth verdict predicts no gain in any dataset. (c) The gain per backbone and which readout collects it.

(c) Best zero-cost metric per backbone: few-shot advantage over Euclidean, mean over the four hierarchical datasets, and which metric collects it (either: within 0.15pp).

| model    | best—R (pp) | metric |
|----------|-------------|--------|
| ViT-T    | +0.58       | either |
| ViT-S    | +0.49       | either |
| ViT-B    | +0.51       | either |
| ViT-L    | +0.87       | either |
| DINOv2-S | +0.68       | H      |
| DINOv2-B | +1.17       | cos    |
| DINOv2-L | +1.84       | cos    |
| DINOv2-G | +2.54       | cos    |
| CLIP-B   | +0.67       | H      |
| CLIP-L   | +0.65       | H      |

Table 14: (continued)

(d) The same correlations on the raw  $\hat{\delta}_{99.9}$ , the record excess and the depth-test  $z$  (Fisher- $z$  CI95; dm: family-demeaned  $r$ ).

| gain      | dataset   | raw $\hat{\delta}_{99.9}$ |       | record excess        |       | depth $z$            |  |
|-----------|-----------|---------------------------|-------|----------------------|-------|----------------------|--|
|           |           | $r$ [CI95]                | dm    | $r$ [CI95]           | dm    | $r$ [CI95]           |  |
| NC H-R    | ImageNet  | -0.84 [-0.96, -0.45]      | -0.83 | -0.73 [-0.93, -0.19] | -0.61 | -0.28 [-0.77, +0.42] |  |
|           | CIFAR-100 | -0.92 [-0.98, -0.68]      | -0.58 | -0.79 [-0.95, -0.33] | -0.21 | +0.52 [-0.16, +0.87] |  |
|           | CIFAR-10  | -0.63 [-0.90, +0.00]      | -0.31 | -0.67 [-0.91, -0.06] | -0.34 | n/a                  |  |
|           | DTD       | -0.67 [-0.91, -0.07]      | -0.67 | -0.76 [-0.94, -0.26] | -0.60 | n/a                  |  |
| FS H-R    | ImageNet  | -0.86 [-0.97, -0.50]      | -0.56 | -0.21 [-0.74, +0.48] | +0.35 | +0.27 [-0.43, +0.77] |  |
|           | CIFAR-100 | -0.62 [-0.90, +0.02]      | -0.51 | -0.79 [-0.95, -0.31] | -0.80 | -0.26 [-0.76, +0.45] |  |
|           | CIFAR-10  | -0.78 [-0.95, -0.31]      | -0.64 | -0.78 [-0.95, -0.30] | -0.66 | n/a                  |  |
|           | DTD       | -0.87 [-0.97, -0.53]      | -0.94 | -0.98 [-1.00, -0.91] | -0.97 | n/a                  |  |
| FS best-R | ImageNet  | -0.83 [-0.96, -0.43]      | -0.50 | -0.08 [-0.68, +0.58] | +0.47 | +0.13 [-0.54, +0.70] |  |
|           | CIFAR-100 | -0.77 [-0.94, -0.27]      | -0.75 | -0.75 [-0.94, -0.23] | -0.65 | +0.22 [-0.48, +0.75] |  |
|           | CIFAR-10  | -0.63 [-0.90, -0.01]      | -0.55 | -0.62 [-0.90, +0.02] | -0.56 | n/a                  |  |
|           | DTD       | -0.81 [-0.95, -0.36]      | -0.78 | -0.90 [-0.98, -0.63] | -0.84 | n/a                  |  |

Table 14: (continued) (d) The calibrated reading: the raw 99.9th-percentile statistic, the record excess and the depth-test  $z$  of Table 9 as predictors. (e) Validation picking retains the oracle advantage and the zero-validation objective rule modestly beats always-cosine. (f) The gate’s value is avoiding the Poincaré projection on flat-label data; cosine is a safe NC default everywhere. (g) The headline  $t = 1/\sqrt{2} \approx 0.71$  was fixed a priori and is conservative for FS.

(e) Few-shot metric-selection policies on held-out episodes (500 validation / 500 test per cell).

| policy (test advantage over Euclidean, pp) | all 60 cells | hierarchical (40) |
|--------------------------------------------|--------------|-------------------|
| always cosine                              | +0.23        | +0.28             |
| objective rule (zero validation)           | +0.28        | +0.41             |
| $\delta$ -gated objective rule             | +0.08        | +0.13             |
| validation-picked per cell                 | +0.63        | +0.78             |
| test oracle (reference)                    | +0.64        | +0.78             |

Table 14: (continued)

(f) Nearest-centroid metric-selection policies.

| NC policy (advantage over Euclidean, pp) | all 60 | flat (20) | hierarchical (40) |
|------------------------------------------|--------|-----------|-------------------|
| always cosine                            | +0.48  | +0.24     | +0.59             |
| always Poincaré                          | -0.24  | -0.70     | -0.00             |
| gated rule (flat→R)                      | +0.34  | +0.00     | +0.51             |
| $\delta$ -gated rule                     | +0.19  | +0.00     | +0.29             |

Table 14: (continued)

(g) Sensitivity to the projection target  $t$  (36 hierarchical cells).

| target $t$        | 0.25  | 0.35  | 0.41  | 0.45  | 0.50  | 0.58  | 0.71  | 0.95  |
|-------------------|-------|-------|-------|-------|-------|-------|-------|-------|
| FS adv. mean (pp) | +0.88 | +0.87 | +0.87 | +0.86 | +0.84 | +0.82 | +0.76 | +0.56 |
| NC adv. mean (pp) | -0.30 | -0.25 | -0.23 | -0.21 | -0.17 | -0.12 | -0.01 | +0.15 |
| % cells FS > 0    | 97    | 100   | 100   | 100   | 100   | 100   | 100   | 100   |

Table 14: (continued)

## B.13 PROVENANCE

| table | generated file        | result file(s)                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5     | tab_q01_census_final  | expR75_census_centered_haar.csv<br>expR52_census_haar_p999_200.csv<br>expR54_census_haar_sup_200.csv expR40b_p999census200.csv<br>expR39c_census200_cache.csv<br>expR57_census_cosine_haar_p999_200.csv<br>expR57_text_cosine_haar_p999_200.csv<br>expl0_local_vs_global.csv expR45_convnet_rows.csv<br>exp20_null_ztable.csv                                                                                                       |
| 7     | tab_q02_text          | expR53_text_haar_p999_200.csv expR53_text_haar_sup_200.csv<br>expR53_text_gauss_p999_200.csv expR48b_text_census200_bsl.csv<br>expR57_text_cosine_haar_p999_200.csv expl8_text_nulls.csv<br>exp4_prompt_variation.csv expl7_wordnet_prompts.csv<br>expR49_template_nulls_bsl.csv expR47_extraction_variance.csv                                                                                                                     |
| 8     | tab_q03_sample_final  | expR62_samplelevel_record.csv expR78_khrulkov_replication.csv<br>expR78_khrulkov_replication_summary.csv                                                                                                                                                                                                                                                                                                                            |
| 9     | tab_q04_depth_final   | expR56_depth_variants.csv expR69_depth_haarhubs_summary.csv<br>expR64b_wn30bal_summary.csv expR70_inet1k_supervised.csv<br>expR52_census_haar_p999_200.csv expR63_meru_record.csv<br>expR65_hier_finetune.csv expR74_decoupling_summary.csv<br>expR71_meru_radial.csv expR77_positive_control.csv<br>expR77_positive_control_verdict.json<br>expR82_radial_control.csv expR82_radial_control_summary.csv                            |
| 10    | tab_q05_power_final   | expR55_depth_power.csv expR55b_depth_power_leafframe.csv<br>expR64b_wn30.csv expR64b_wn30_summary.csv<br>expR64b_wn30bal_summary.csv expR67_frame_choice.json<br>expR69_depth_haarhubs_summary.csv<br>expR80_implanted_alignment.csv expR80_decision.csv<br>expR79_synthetic_deep_poincare.csv<br>expR81_deep_per_backbone.csv<br>expR81_deep_per_backbone_summary.csv expR83_flat_balanced.csv<br>expR83_flat_balanced_summary.csv |
| 11    | tab_q06_treemap       | exp22_tree_similarity_imagenet.npz exp23_treemap_controls.npz<br>exp23_config_diagnostics.json<br>expR58_treemap_cutfree_summary.csv expR84_tree_ceiling.csv<br>expR84_tree_ceiling_summary.csv                                                                                                                                                                                                                                     |
| 12    | tab_q07_wordnet_final | exp3_alignment.csv exp28_recovery_per_config.csv<br>exp8_pl_recovery.csv expl_delta_controls.csv<br>exp8_p6_pooling.csv expR70_inet1k_supervised.csv<br>expl4_dbpedia.csv expR61_dbpedia_record.csv<br>exp27_dbpedia_treemap.json expl6_hierarcaps.csv                                                                                                                                                                              |
| 3     | tab_q08_robust_final  | expR72_budget_record.csv expR72_budget_record_summary.csv<br>expR59_imagenet_bootstrap_summary.csv<br>expR73_transfer_bootstrap_record_summary.csv<br>expR66_joint_sensitivity_summary.csv<br>expR60_c_sweep_record.csv analysis4_finetuning.csv<br>el_delta_by_layer.csv expR66c_joint_sensitivity_summary.csv                                                                                                                     |
| 14    | tab_q09_corollary     | table1_regenerated.csv exp2_metric_controls.csv<br>exp2b_normalized_stack.csv expl3_mcnemar.csv<br>night/correlation_cis.csv exp20_null_ztable.csv<br>expR68_corollary_excess.csv exp24_val_metric_selection.csv<br>night/t_sweep.csv                                                                                                                                                                                               |
| 4     | tab_q10_calibration   | expl_delta_controls.csv expR42_star_calibration.csv                                                                                                                                                                                                                                                                                                                                                                                 |
| 13    | tab_q13_local         | exp21b_local_global_K200.csv                                                                                                                                                                                                                                                                                                                                                                                                        |
| 6     | tab_q14_panel         | (model panel: stated in the generator; parameter counts<br>from the model cards)                                                                                                                                                                                                                                                                                                                                                    |

Table 15: Provenance index: every appendix table is generated by script from the released per-cell result files listed here; no table value is hand-transcribed. The same mapping for main-text tables and figures is annotated in the source and shipped with the code release.